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CONFINED DIFFUSION OF NANOPARTICLES USING iSCAT MICROSCOPY

by

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Layman Summary

Many products are made using very tiny pieces of material called nanoparticles. These are found in things like plastics, medicines, and food. To make sure these products are safe and good quality, we need to measure what these tiny pieces are like. For example, their size and how they behave.

However, nanoparticles are so small that we cannot see them clearly, even with a microscope that uses light. So we need to find new ways to study them.

This thesis develops an alternative method to do this. It measures where a particle is located in space, and it watches how that position changes over time. Using mathematics, we can then learn more about the particle from the way it moves. This helps us measure nanoparticles more accurately.

Abstract

The understanding of thermal movement of paricles within a confined space has been studied extensively in order to understand biological, chemical, and physical processes. Although the theoretical understanding of this confined diffusion is well defined in literature, not much work has been done to understnad the effects of electrostatic and hydrodynamic contributions to the size measurements measured from the motion of nanoparticles, as measurements are often instrinsically performed under confinement. This is due to the fact that 3D particle-tracking is not as commonly used as other, higher throughput sizing techniques. By employing developed techniques of nanoparticle tracking in tandem with a full theoretical model, the non-trivial effects of confinement on size-diffusivity were directly investigated and characterized. The work described in the thesis advances the method and tools of 3D nanoparticle analysis both theoretically and in application.

The first section of the thesis describes a theoretical model of Brownian motion based on the confined-Langevin model using a Debye-Huckel electrostatic potential. The model was implemented, based on literature [1], into a computer-based Python simulation in order to parameterize and create prior distributions of particle motion.

The second section describes a modified method of 3D tracking to obtain faster calculations of particle position from optical microscopy images. This was based on an algorithm described in literature [2]. The validation of the method was done on exact point-spread function simulation data with an accuracy of 1 nm.

The third section describes the aquisition of experimental data using a custom interferometric scattering microscope (iSCAT) with image pixel resolution 55 nm aquiring images between 400 and 1000 frames per second. The data consisted of images of scattering fringes from nanoparticle species and was analyzed using the algorithm developed in section 2. The resultant information was accuracte 3D particle tracks. The models developed in sectino 1 were applied to the tracks to obtain infer particle size and charge.

The thesis presents a method of finding particle size, motion, and electrostatics which can present additional tools to analyze nanoparticle characterization, which was the goal of the thesis.

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Contents

Nomenclature **5**

1 Introduction **7**

2 Brownian Motion **10**

2.1 Classical Description of Einstein's Brownian Motion 10
2.2 Langevin Dynamics 12
2.3 From Master Equation to Fokker-Planck 13
2.4 Confined Diffusion Processes 14
2.5 Electrostatic Potentials 14
2.6 Equilibrium Distributions for Confined Systems 16

3 Interferometric Scattering Principles **18**

3.1 Light as a casual field 18
3.2 Electromagnetic Scattering of of Homogenous Spheres 19
3.3 Rayleigh Scattering 20
3.4 Scattering Microscopy 21
3.5 Interferometric Light Scattering (iSCAT) 22
3.6 Resolution limits of optical microscopy 23
3.7 Sources of noise in imaging 24

4 Biological Application **27**

4.1 Diffusion in cells 27
4.2 Brownian motion as a method for measurements of size in biology 28
4.3 Lipid Nanoparticles 28
4.4 LNP topology types 29

5 Langevin Simulations **30**

5.1 Simulation Equations 30
5.2 Surface Charge Estimator 31
5.3 Results 32

5.3.1 Mean-Squared Displacement in Confinement . . . 32
5.3.2 Surface Charge estimation . . . 35
5.4 Conclusion . . . 36

6 iSCAT z-estimation Simulations 37
6.1 Method Overview . . . 37
6.2 Raw image background subtraction . . . 38
6.3 iSCAT (x, y) Tracking . . . 39
6.4 Angular Integral . . . 41
6.5 Nuisance parameter Estimation . . . 41
6.6 2D Chain-link . . . 43
6.7 Simulated iPSF localization . . . 44
6.8 3-parameter Estimation . . . 45
6.8.1 Grid-Search Estimation . . . 46
6.9 Conclusions . . . 46

7 3D confinement with iSCAT 47
7.1 Methods . . . 47
7.1.1 Aparatus schematic . . . 47
7.1.2 Cover-Glass . . . 48
7.1.3 Convex-lens induced confinement . . . 48
7.1.4 Sample Details . . . 49
7.1.5 Analysis Method . . . 49
7.2 Results . . . 49
7.2.1 Polystyrene PBS series . . . 50
7.2.2 Axial Diffusivity . . . 50
7.2.3 LNP Diffusivity . . . 51
7.3 Discussion of Results . . . 52

8 Conclusion 53
8.1 Conclusion . . . 53
8.2 Future outlook . . . 53

A Surface Charge Estimator Derivation 57
A.1 $\hat{B}$ MLE . . . 57
A.2 $\mathbb{V}[\hat{B}]$ . . . 58

B Axial Estimation Algorithm Pseudocode 59

Nomenclature

Symbol	Description
δ_{ij}	Kronecker delta
$\delta(\cdot)$	Dirac delta distribution
$\vec{x}$	Position vector
$\vec{v}$	Velocity vector
m	Mass
t	Time
κ	Inverse Debye length
ζ_p	Particle zeta potential
ψ_w	Wall zeta potential
Γ_i	Stern layer of species i
Γ	Fluctuation–dissipation magnitude
j	Probability current
W_t	Gaussian white-noise process
σ_w	Surface charge density
D	Diffusion coefficient
h	Length
Ψ	Electrostatic potential
ρ	Volumetric charge density
ε_0	Permittivity of free space
ε_r	Relative permittivity of the material
Z	Ion number (valence)
C	Concentration

Symbol	**Description**
β	$1/k_bT$
e_0	Elementary charge
a	Spherical particle radius
η	Fluid viscosity
k_b	Boltzmann constant
T	Temperature
$\mu_\parallel$	Parallel (axial) mobility coefficient
$\mu_\perp$	Perpendicular (transverse) mobility coefficient
Re	Reynolds number

Chapter 1

Introduction

In certain biological systems, particle size is proportional to the functional properties of the particle. For example, mRNA-lipid nanoparticles (mRNA-LNP) based medicines are expressed depending on their endosomal escape efficiency, a highly sensitive size-selective process [3–7]. In the context of pharmacological production, the control of particle size is important to ensure drug potency and efficiency.

In the limit of microscopic objects smaller than the wavelength of light, such as particles $\sim 150\,\text{nm}$, the size becomes un-resolvable by the light alone [8, 9]. In order to measure the size, one must use alternative approaches.

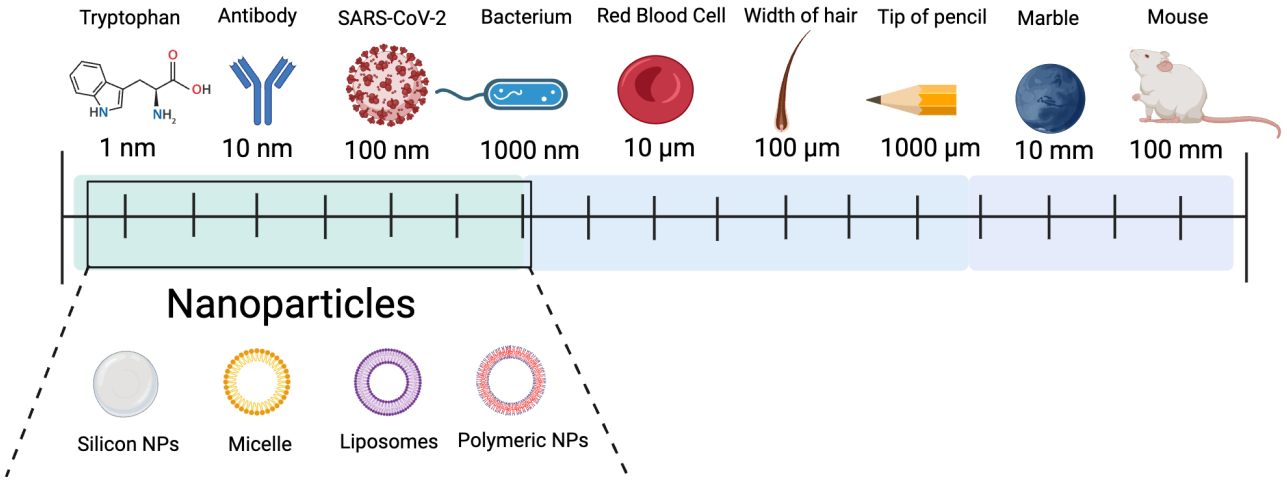


Figure 1.1: Size comparison of various particles. Nanoparticles are defined as particles where the largest dimension is at most $\sim 200\,\text{nm}$ REF.

One universal property of sub-micron particles is that they undergo an apparent size-related stochastic motion when suspended in media [10–12]. At the nano-scale, thermal energy, k_bT, is comparable to, or larger than, the gravitational and inertial energy scales that typical govern macroscopic dynamics. In the case of continuous media made of atomic subunits, the thermal

energy emerges into a motion when the media's Reynold's numbers is much lower than one [12],

$$\mathrm{Re} \ll 1.$$

In this environment, the media's viscous drag dominates over the inertial force, which constitutes a memoryless, random motion [1].

The relation between the particle's size and its motion is explained in Einstein's seminal paper [11] where the rate of transport, or diffusion, is related to the particle's hydrodynamic radius,

$$D(\vec{x}, a) = \frac{k_b T}{\gamma(\vec{x}, a)}, \tag{1.1}$$

where $\gamma(\vec{x}, a)$ is the viscous drag at the position $\vec{x}$ of a particle of hydrodynamic radius a. In the case of a spherical paricle in an incompressible fluid, $\gamma(a) = 6\pi\eta a$, where η is the fluid viscosity.

In order for the size estimate obtained from stochastic particle trajecories to be statistically relevant, the number of data points in the positional trajecory must be sufficient. Since the statistical uncertainty of the diffusion coefficient scales as $\sim \sqrt{N}$, a higher number of particle locations yield a more precise size inference [13].

Long particle trajectories are typically obtained using imaging the particle position over time using optical microscopy. In optical measurements, the long-particle trajectory requirement favors the use of a scattering microscopy over fluorescence microscopy. Fluorescent microscopy utilizes excited fluorophores to produce signal. These fluorophores have finite lifetimes, to which they oxidize and stop producing light, known as photobleaching [14]. In addition, fluorescent microscopes, although possible to obtain three-dimensional position, are often only accurate for two-dimensional localization REF.

In order to obtain large number of particle positions, while maintaining independent particle specificity, particles are often spatially confined within the field of view of an optical imaging system for an extended period of time REF?. Without this axial confinements, particles rapidly diffuse out of the focal plane. In prior work, convex-lens induced confinement (CLiC) measurements have typically utilized micro-apertures, $\sim 1.5\,\mu\mathrm{m}^3$, in order to confine single particles for temporally extended measurements via fluorescence microscopy [15–17]. In complement, Nanoparticle tracking analysis (NTA) measurements follow the same principles, where flow is applied to particles confined between two plates and are imaged using side-scattering microscopy [18]. in NTA, however, particles are still able to diffuse in and out of the focal volume.

Although experimentally feasible, confinement intrisicaly modifies the size-diffusivity relation rendering it non-trivial by introducing a position dependent viscous drag. This non-trivial behavior can been shown to alter the diffusivity of a particle by a significant amount [12, 19].

[1] This is also a function of temporal resolution [12].

Since the diffusivity of a particle in confinement is a function of position, in order to recover the true size of a particle diffusing, it is necessary to have the full spatial trajectory of the particle's motion with respect to the confinement barriers. As such, two dimensional tracking alone is insufficient; without the axial dimensions the spatially dependent correction cannot be inferred. Inverting the size-diffusivity relation demands the full three-dimensional trajectory, in particular the axial distance to the wall.

In order to obtain an optical trace of three-dimensional scattering, interference methods, such as interferometric light-scattering microscopy, are highly favored as typical fluroescence measurements lack an extended temporal measurement of three-dimensional position. In addition, many commonly used optical methods are limited to only two-dimensional slices of a focal plane, which is typically on the order of a few hundred nanometers.

Thus, this thesis establishes a three-dimensional tracking method of confined diffusion and builds on applications of measurement of the size-diffusivity relationship, where the z-motion is used to evaluate the position-dependent confinement as well as providing complementary particle size information that is independent on the media viscosity. The thesis consists of 8 chapters. Chapter 2 provides a description of the literature of confined brownian motion over the last century. Chapter 3 describes the theoretical description of interferometric scattering microscopy (iSCAT) and physical limits that apply in optical theory. Chapter 4 discusses the application of Brownian motion and iSCAT in a biological context, which details regarding lipid-nanoparticles and why they need to be studied on the single-molecule level. Chapter 5 consists of investigations of confined diffusion using Langevin simulations. A simulated system was developed and the size-diffusivity relation in an electrostatically screened system was investigated. Chapter 6 describes the 3D iSCAT tracking technique developed during this work. The method overview and validation and explicitly developed and validated Chapter 7 describes the method applied to experimental data and utilizes various nanoparticle species, highlighting their varied diffusions. Chapter 8 concludes the world and provides a future outlook on the topic and steps which need to be advanced.

Chapter 2

Brownian Motion

For particles smaller than the diffraction limit, where optical theory is discussed in 3, thermal motion can be used as a physical *ruler*. The observation of this motion originated around the year 1827 when Robert Brown observed a stochastic motion of clay particles originating from plant pollen under his microscope [10]. This stochastic motion is known as *Brownian Motion* and is invertibly related to the size of the particle.

This chapter aims to establish the theoretical background underlying the relationships between the diffusivity, particle size, and spatial coordinate. This process requires sufficient mathematical detail as no single displacement carries enough information to define the whole trace, however, as one observes this motion for a longer period of time, the randomness tends to converge to a lawful, physical origin.

This chapter introduces the mathematics of Brownian motion and diffusion-type processes. Einstein's diffusion- equation argument and Langevin's stochastic equation of motion are discussed and shown to be equivalent through the Fokker-Planck formalism. In the case of diffusion under confinement, where hydrodynamic wall effects modify the mobility and electrostatic interactions, described by the Debye-Hückle interactions, govern particle-wall interactions in ionic solution. Finally, the subject of equilibrium distributions are discussed with regards to spatial probability distributions arising in confined diffusion processes.

2.1 Classical Description of Einstein's Brownian Motion

The physical phenomenon of diffusion can be rigorously described as the emergent behavior atomic particle undergoing random collisions with members of the surrounding environment. In the microscopic limit, where the radius of a particle is sufficiently small and thermal fluctuations dominate its dynamics, this erratic motion is referred to as Brownian motion, or more colloquially, a diffusive process [10].

The classical description of Brownian motion is concerned with how the molecular-kinetic theory of heat relates with microscopic particle suspensions in liquid or gas. In particular, how do these particles perform movements such that they can be viewed by a microscope? Einstein's argument is one of macroscopic motion. Mathematically, one can use a Taylor series expansion over a general concentration function to show a natural property of the system. Using Einstein's original notation, suppose that $\phi(\Delta)$ denotes the probability to find a particle in some space Δ subject to the normalization

$$\int \phi(\Delta) d\Delta = 1 \tag{2.1}$$

Assume that the medium is isotropic such that ϕ is even, $\phi(\Delta) = \phi(-\Delta)$.

Let $f(x,t)$ denote some number of particles at the point x. Einstein takes a finite slice of this volume and equates it to a probability space;

$$f(x, t+\tau) = \int f(x+\Delta, t)\phi(\Delta) d\Delta \tag{2.2}$$

The argument here is that we can taylor expand both sides of this equation,

$$f(x, t+\tau) \approx f(x,t) + \tau \frac{\partial f(x,t)}{\partial t} \tag{2.3}$$

And in space;

$$f(x+\Delta, t) \approx f(x,t) + \Delta \frac{\partial f(x,t)}{\partial x} + \frac{\Delta^2}{2} \frac{\partial^2 f(x,t)}{\partial x^2} \tag{2.4}$$

Cancelling terms, one finds the classical diffusion equation to nearest order;

$$\frac{\partial f(x,t)}{\partial t} = \underbrace{\left[\frac{1}{\tau} \int \frac{\Delta^2}{2} \phi(\Delta) d\Delta \right]}_{D} \frac{\partial^2 f(x,t)}{\partial x^2} \tag{2.5}$$

Where we typically associate the diffusion coefficient with the second moment of the probability distribution

$$D = \frac{1}{\tau} \int \frac{\Delta^2}{2} \phi(\Delta) d\Delta \tag{2.6}$$

The friction coefficient here is related to the solution of the Navier-Stokes equations at a low Reynolds number. Specifically, when there are no inertial forces and $\mathrm{Re} \ll 1$, then the drag force on a particle,

$$\vec{F}_d = 6\pi\eta a v, \tag{2.7}$$

where $\gamma_0 = 6\pi\eta a$, η is the fluid viscosity, and a is the spherical particle's radius. This relation can be directly applied to the form of the diffusion coefficient to recover Einstein's famous formula;

$$D = \frac{k_b T}{6\pi\eta a}. \tag{2.8}$$

2.2 Langevin Dynamics

In typical experiments, such as Robert Brown's particulates or diffusing lipid-nanoparticles, the observed Brownian particle move within a fluid volume. If one assumes that this particle exists in the thermodynamic equilibrium, it should obey the equipartion theorem. However, the observed motion appears to have more energy than predicted by the equipartion at certain times, indicating that an additional contribution to the dynamics must be considered REF.

This missing component of motion can be described by the bombardment of the surrounding particles in the liquid. In particular, the system experiences a *random* force on it at every time step.

$$m\ddot{\vec{x}} = -\frac{\dot{\vec{x}}}{B} + \vec{F}(t) \tag{2.9}$$

where B is the mobility (drag term is $-\frac{\dot{\vec{x}}}{B}$), which is represented in a viscous friction term, and $\vec{F}(t)$ is a stochastic force term, meaning it takes a random value from some distribution f for every time step t. This model, although easily written, is quite profound and its analytical treatment is non-trivial. Here, $x(t)$ is a stochastic process, where each realization of $F(t)$ produces a unique trajectory of $x(t)$.

The statistical content of $\vec{F}(t)$ is determined by the moments and auto-correlation. Notably, this relation is physically described by the fluctuation-dissipation theory, which relates the drag force to the random force. In Brownian motion, this takes the form

$$\langle \vec{F}(t) \rangle = 0, \quad \langle \vec{F}_i(t)\vec{F}_j(t') \rangle = \Gamma\delta_{ij}\delta(t-t') \tag{2.10}$$

Which stands in as delta-correlated noise, or that of a Wiener process. Notably, Γ is constrained by the equipartion theorem. This can be shown by calculating the velocity autocorrelation function (VACF) and choosing the large t limit to agree with the equipartion theorem in 3-dimensions. This fixes the fluctuation constant to being

$$\Gamma = \frac{2k_b T}{B}. \tag{2.11}$$

One can take the ensemble average of this device, and notably, extract correlation information. Rewriting in terms of velocity, and assuming zero drift velocity one finds

$$m\dot{\vec{v}} = -\frac{\vec{v}}{B} + \vec{A}(t), \quad \int_{-\infty}^{\infty} A(t)dt = 0. \tag{2.12}$$

This represents a linear differential equation with a solution hat defines the velocity autocorrelation function. This solution is

$$\langle \vec{v}(0) \cdot \vec{v}(t) \rangle = \frac{3k_bT}{m} e^{-t/(mB)}. \tag{2.13}$$

Furthermore, when one averages the product of the above equation with $\vec{x}$ and integrates over time, we find the following form of the mean-square displacement. Note that $\langle v^2 \rangle = \frac{3k_bT}{m}$, then the solutions of the position mean-squared displacement are written as

$$\begin{aligned} \langle x^2 \rangle &= 6k_bTmB^2 \left[\frac{t}{mB} - (1 - e^{-t/(mB)}) \right] \\ &\begin{cases} \xrightarrow{t \ll mB} \frac{3k_bTt^2}{m} & = \langle v^2 \rangle t^2 \\ \xrightarrow{t \gg mB} \frac{6k_bTmB}{m} t & = (6Bk_bT)t \end{cases} . \end{aligned} \tag{2.14}$$

Letting $D = Bk_bT$, we recover Einstein's relation with ease.

The unique advantage of the Langevin Formulation is that it is readily simulate-able [1]. It can be shown that using an Euler-Maruyama scheme, the stochastic differential equation can be integrated over realizations of $F(t)$ using a random number generator, recovering the dynamics of an ideal brownian system [20].

2.3 From Master Equation to Fokker-Planck

To show that the two methods are equal, we can use the typical approach to equilibrium provided by the master equation. Suppose that $f(x,t)dx$ representation that he probability of an arbitrary particle in the ensemble has a displacement between x and $x + dx$. We find that if f is a probability distribution, we can write down the master equation, let W represent the traditional probability transition rate;

$$\frac{\partial f(x,t)}{\partial t} = \int_{-\infty}^{\infty} \left[f(x',t)W(x'|x) - f(x,t)W(x|x') \right] dx' \tag{2.15}$$

This structure is a statement about the probability of positional transitions, which is fundamentally equivalent to the Langevin equation[1] The right hand side can be expanded using the Kramer-Moyal expansion and truncated at second order, yielding the Fokker-Planck Equation.,

$$\frac{\partial f(x,t)}{\partial t} = -\frac{\partial}{\partial x} \left[\sigma(x) f(x,t) \right] + \frac{1}{2} \frac{\partial^2}{\partial x^2} \left[D(x) f(x,t) \right], \tag{2.16}$$

[1] Given a Markovian system.

where the diffusive and drift terms are parameterized over the transition probabilities,

$$\begin{aligned} \sigma(x) &= \int_{-\infty}^{\infty} \xi W(x|\xi) d\xi \\ D(x) &= \int_{-\infty}^{\infty} \xi^2 W(x|\xi) d\xi. \end{aligned} \tag{2.17}$$

In the case of zero drift, we can clearly see that this form equates to the diffusion equation of the Langevin formulation and the Einstein-ian formulation are the same. This important property allows us to investigate solutions and dynamics of the two systems interchangeably. In particular, relating expected results from simulation-based behavior of Langevin problems.

2.4 Confined Diffusion Processes

The effect of diffusion near an interface on a nearby particle is quite particular. The effect on confinement can be first imagined as the effects from the fluid friction flow. Notably, the particle being immersed in fluid invites a stokes friction which effects the perpendicular and parallel flows. Hiding Faxén derived the relations in no-slip boundary conditions near a brownian particle;

$$\begin{aligned} \mu_{\parallel}\left(\frac{a}{h}\right) &\approx \mu_{\infty}\left[1 - \frac{9}{16}\left(\frac{a}{h}\right) + \frac{1}{8}\left(\frac{a}{h}\right)^3 - \frac{45}{256}\left(\frac{a}{h}\right)^4 - \frac{1}{16}\left(\frac{a}{h}\right)^5\right] \\ \mu_{\perp}\left(\frac{a}{h}\right) &\approx \mu_{\infty}\left[1 - \frac{9}{8}\left(\frac{a}{h}\right) + \frac{1}{2}\left(\frac{a}{h}\right)^3\right] \end{aligned} \tag{2.18}$$

In the case of a two-wall confinement, the form of the diffusion coefficient has a lot of depth in wealth. In application, it has been shown that the form of the Onseen drag in terms of mobility coefficients.

$$\gamma_2(z) \approx \gamma_0 + [\gamma_1(z) - \gamma_0] + [\gamma_1(H - z) - \gamma_0] \tag{2.19}$$

2.5 Electrostatic Potentials

A particle diffusing near a charged wall in an electrolyte will feel a screened electrostatic interaction. The Debye-Huckel theory is a description of the electromagnetic interactions between ions in a solution of ionic compounds. The potential in a field of ions must be solved in terms of electrostatics in a media. This is described by Poisson's equation

$$\nabla^2 \Psi(\vec{x}) = -\frac{\rho(\vec{x})}{\varepsilon_0 \varepsilon_r} \tag{2.20}$$

In a sea of ions, we must carefully define the charge density as it pertains to the system. In a general solution of ions, there will be j species of ions.

$$\nabla^2\Psi(\vec{x}) = -4\pi\frac{\rho_j(\vec{x})}{\varepsilon}$$

The charge distribution of a single ion concentration is essentially the density of the ionic positions. It would be tedious and unfeasible to count every ion's position exactly as we are assuming too many things. Rather, one can use a statistical ensemble of ions via a canonical ensemble, which is a good approximation here.

$$\rho_i(\vec{x}) = \sum_i C_i e_0 Z_i \exp\left[-\beta Z_i e_0 \Psi(\vec{x})\right] \tag{2.21}$$

Here the energy of the system is defined as the potential energy scaled by the charge defined as $U = qV$. The full equation is known as the Poisson-Boltzmann equation, where $\Psi(\mathbf{r})$ is the potential for all space.

$$\nabla^2\Psi(\vec{x}) = -\frac{1}{\varepsilon_0\varepsilon_r}\sum_{i=1}^{\infty} C_i e_0 Z_i \exp\left[-\beta Z_i e_0 \Psi(\vec{x})\right] \tag{2.22}$$

There is a nice approximation we can make given that the potential of the solution is small. We linearize the exponential to first order here.

$$\begin{aligned}
\nabla^2\Psi(\vec{x}) &\approx -\frac{1}{\varepsilon_0\varepsilon_r}\sum_{i=1}^{\infty} C_i e_0 Z_i \left[1 - \beta Z_i e_0 \Psi(\vec{x})\right] \\
&= -\frac{1}{\varepsilon_0\varepsilon_r}\sum_{i=1}^{\infty} \left[C_i e_0 Z_i - \beta C_i Z_i^2 e_0^2 \Psi(\vec{x})\right] \\
&= -\sum_{i=1}^{\infty}\left[\frac{C_i e_0 Z_i}{\varepsilon_0\varepsilon_r} - \kappa_i^2\Psi(\vec{x})\right]
\end{aligned}$$

Where κ_i^2 is related to the debye-length, a unit of the electrostatic screening caused by the ionic interactions. The Debye length is defined as $\lambda_D = \kappa^{-1}$.

When the bulk solution is electro-neutral, $\sum_i C_i Z_i = 0$, then the zeroth therm can cleanly drop out, leaving only the approximation;

$$\nabla^2\Psi(\vec{x}) = \kappa^2\Psi(\vec{x}), \quad \kappa_i^2 = \frac{\beta e_0^2}{\varepsilon_0\varepsilon_r}\sum_i C_i Z_i^2. \tag{2.23}$$

This equation has an obvious solution which are the real exponential functions;

$$\Psi(\vec{x}) = Ae^{-\kappa\vec{x}} + Be^{\kappa\vec{x}}. \tag{2.24}$$

The pre-factor A, B is the strength of the electric potential. In the case of ionic potentials, the strength of the pre-factor is a number of the number of ions and their charge density surrounding the surface. Thus, one can parameterize this pre-factor based on the geometry.

In the approximations done in this thesis, we assume a linear symmetric potential, as well as a spherical particle between them. The form of this potential is given by

$$B = C\Gamma_p\Gamma_w, \tag{2.25}$$

where $|\psi|$ is a fixed amplitude due to charge density, and Γ_w, Γ_p are terms associated with the Stern layers of the wall and the particle;

$$\begin{aligned} C &= \frac{64\pi\varepsilon_0\varepsilon_r a k_b T}{e_0^2} \\ \Gamma_w &= \tanh\left(\frac{e\psi_w}{4k_bT}\right) \\ \Gamma_p &= \tanh\left(\frac{e\zeta_p}{4k_bT}\right) \\ \psi_w &= \frac{2k_bT}{e}\text{asinh}\frac{e\lambda_D\sigma_w}{2\varepsilon_0\varepsilon_r k_b T}. \end{aligned} \tag{2.26}$$

Note that ψ_w is the Stern layer of the wall related to the surface charge density of the wall [21]. This is chosen for a more intuitive measurable of a wall-like structure.

2.6 Equilibrium Distributions for Confined Systems

In the case of interfaces, we have boundary conditions that case the particle to reflect off of the sides. Notably, the probability flux is set to 0. This implies that $\partial_x j = 0$ as well. Then, we need to solve the adjusted Fokker-Plank equation;

$$j = -\frac{\partial}{\partial x}\left[D(x)f(x)\right] - \nabla V(x)f(x) \tag{2.27}$$

Setting j to 0, we find that the solution of this equation defines the equilibrium distribution after a long time.

$$f(x) = \frac{\exp\left[-\beta\int_0^x(\frac{\nabla V(y)}{D(y)})dy\right]}{\exp\left[-\beta\int_{-H}^{H}(\frac{\nabla V(y)}{D(y)})dy\right]} \tag{2.28}$$

This equation defines the equilibrium distribution of the brownian diffusion after a long time.

Chapter 3

Interferometric Scattering Principles

To precisely determine the size of nanoscale particles,it is essential to obtain long and procise positional trajectories. To achieve this, a particle localization method that maintains a consistent signal over extended measurement periods is required. One method that enables such localization of particle traces is interferometric scattering microscopy (iSCAT) [14].

iSCAT is a relatively modern optical imaging technique that utilizes light scattering contrast to localize particle signal. The particle's transverse position can be directly rendered by utilizing the scattering pattern's angular symmetry. Similarly, utilizing the phase difference in the scattering pattern between series of images the relative axial distance can be measured. Thus, the 3D positional information is encoded in the scattering pattern in iSCAT.

The phase information stored in iSCAT images contain relative axial position which is more consistent and reliable relative to other techniques REF.

We begin with the basics of relating light to electromagnetism, starting with the Maxwell and Helmholtz relations. We continue discussing the scattering of a various size range of spheres, which is the approximation we take in this thesis. We then discuss the various aberrations and Considerations before introducing iSCAT microscopy. We tease the experimental apparatus used in this work and compare the method to other methods currently used in the field. Finally, the electromagnetic theory of scattering off spheres, discussion of general optical scattering microscopy theory, and principle of iSCAT will be discussed.

3.1 Light as a casual field

In order to discuss the spatial distribution of light scattering, it is necessary to use a framework to develop the fundamental properties. On the quantum level, electromagnetic wave field consist of photons [22]. This fact motivates the use of the theory of electromagnetic waves, in particular Maxwell's equations, to be applied to optical field. The optical field is defined with respect to

light intensity, which is defined as the modulus of itself,

$$I = |\vec{E}(\vec{x}, t)|^2, \tag{3.1}$$

where $\vec{E}(\vec{x}, t)$ is defined as the optical field[1]. This principle relation allows us to form models using electromagnetic theory to map out how the optical fields behave.

3.2 Electromagnetic Scattering of of Homogenous Spheres

In iSCAT, the light collected at the camera is originated from light scattering off of sphere-like particles and mapped directly to axial position. To simplify the mathematics and obtain positional information from such scattering fields, the scattering dynamics can be modelled as spheres.

The optical theory of light will obey the Maxwell's equations, which motivates solutions to obey the wave equation for the electric and magnetic Fields

$$\nabla^2 \vec{E} + k^2 \vec{E} = 0, \quad \vec{H} = \frac{1}{i\omega\mu_0} \nabla \times \vec{E}. \tag{3.2}$$

The eigenfunctions of the wave equation are the harmonics of the Laplacian operator, ∇^2, in an arbitrary coordinate basis. For isotropic scattering, radial scattering, the spherical basis has been used in literature [24]. The solutions of the vector-wave equation are known as the vector spherical harmonics.

Let ψ be a solution to the scalar wave equation,

$$\nabla^2 \psi + k^2 \psi = 0, \tag{3.3}$$

then the generator of the general solution is taken by sum of components. This the solution is given by the separable form[25],

$$\begin{aligned} \psi(r, \theta, \phi) &= \Phi(\phi)\Theta(\theta)R(r), \\ \Phi(\phi) &\in \{\cos(m\phi), \sin(m\phi)\}, \\ \Theta(\theta) &= P_n^m(\cos\theta), \\ R(r) &\in \{j_n, y_n, j_n + iy_n, j_n - iy_n\}. \end{aligned} \tag{3.4}$$

Here, the symbols j_n, y_n represent the spherical Bessel functions. Their complex sums, $j_n \pm iy_n$, are the hankel functions. The vector spherical harmonics are best described by separating these solutions into even and odd solutions as their pre-factor will be cos or sin respectively, found in

[1] As opposed to an electric field, where $\vec{E}_{\text{optic}}(\vec{x}, t) = \sqrt{\varepsilon_0 n_m c/2}\vec{E}_{\text{elec}}(\vec{x}, t)$. The normalization constant affects how the fields relate to the measured light intensity [23]

Eqns 4.17 to 4.20 in Bohren [24].

$$\begin{aligned} \vec{M}_{emn} &= \vec{\nabla} \times (\vec{r}\psi_{emn}), \quad \vec{M}_{omn} = \vec{\nabla} \times (\vec{r}\psi_{omn}), \\ \vec{N}_{emn} &= \frac{1}{k}\vec{\nabla} \times \vec{M}_{emn}, \quad \vec{N}_{omn} = \frac{1}{k}\vec{\nabla} \times \vec{M}_{omn.} \end{aligned} \tag{3.5}$$

A scattering solution will consist of incident and scattering light. Here, Tte incident light can be assumed to be a polarized plane wave along the x axis without loss of generality. The general form of these equation are then the sum of the spherical vector harmonics. When subject to the energy conservation boundary condition,

$$(\vec{E}_i + \vec{E}_s - \vec{E}_1) \times \hat{r} = (\vec{H}_i + \vec{H}_s - \vec{H}_1) \times \hat{r} = 0, \tag{3.6}$$

Utilizing the orthogonality of the spherical vector harmonics, one arrives are the exact solution known as the Mie Theory form of scattering off of a sphere [24],

$$\begin{aligned} \vec{E}_i &= E_0 e^{ikr\cos\theta}\hat{x} \\ \vec{E}_s &= \sum_{n=1}^{\infty} E_n(ia_n \vec{N^{(1)}}_{e1n} - b_n \vec{M^{(1)}}_{o1n}), \\ \vec{E}_1 &= \sum_{n=1}^{\infty} E_n(ic_n \vec{M^{(1)}}_{o1n} - d_n \vec{N^{(1)}}_{e1n}). \end{aligned} \tag{3.7}$$

In this thesis, only the scattering field is of importance. The amplitude factors of interest are,

$$\begin{aligned} a_n &= \frac{m\Psi(mx)\Psi'(x) - \Psi(x)\Psi'(mx)}{m\Psi(mx)\xi'(x) - \xi(x)\Psi'(mx)}, \\ b_n &= \frac{\Psi(mx)\Psi'(x) - m\Psi(x)\Psi'(mx)}{\Psi(mx)\xi'(x) - m\xi(x)\Psi'(mx)}, \end{aligned} \tag{3.8}$$

where Ψ, ξ are the Riccatti-Bessel functions [24].

3.3 Rayleigh Scattering

In the limit of a small particle from far away, we can mae some cancellations in these constants to simplify the work. Notably, the Hankel function approaches the limit of a spherical wave,

$$h_n(kr) \xrightarrow{kr \gg 1} (-i)^{n+1}\frac{e^{ikr}}{kr} \tag{3.9}$$

In this length-scale, we have that $a_1 \approx \frac{-2i}{3}(ka)^3\frac{m^2-1}{m^2+2}$, and $b_1 \approx 0$. Thus for $n = 1$ we have that

$\pi_1 = 1, \tau_1 = \cos\theta$, thus

$$\begin{aligned} M^{(3)} &\approx (-i)^{n+1}\frac{e^{ikr}}{kr}\left(\cos\phi\pi_n\hat{\theta} - \sin\phi\tau_n\hat{\phi}\right) \\ N^{(3)} &\approx (-i)^{n+1}\frac{e^{ikr}}{kr}\left(\cos\phi\pi_n\hat{\theta} - \sin\phi\tau_n\hat{\phi}\right) \end{aligned} \tag{3.10}$$

can be written in the Rayleigh approximation in the form of the full scattering field,

$$\vec{E}_s = \frac{iE_0e^{ikr}}{kr}\left[\hat{\theta}\cos\phi\sum_{n=1}^{\infty}\frac{2n+1}{n(n+1)}(a_n\tau_n + b_n\pi_n) + \hat{\phi}\sin\phi\sum_{n=1}^{\infty}\frac{2n+1}{n(n+1)}(a_n\pi_n + b_n\tau_n)\right]. \tag{3.11}$$

After subtituting the terms for $n = 1$, there is a neat simplification,

$$\begin{aligned} \vec{E}_{s,\perp} &= \vec{E}_{i,\perp}\frac{k^2\alpha e^{ikr}}{r}, \\ \vec{E}_{s,\parallel} &= \vec{E}_{i,\parallel}\frac{k^2\alpha e^{ikr}}{r}\cos\theta. \end{aligned} \tag{3.12}$$

Singling out the scattering field and noting the form of the wave-vector, $k = \frac{2\pi}{\lambda/n}$, one finds the final scattering field of spheres on the order of the length of the wavelength of the incident light,

$$\vec{E}_s = \vec{E}_0e^{ikr}\frac{\sqrt{8}\alpha\pi^2n_m^2}{\lambda^2r}\sqrt{1+\cos\theta}. \tag{3.13}$$

3.4 Scattering Microscopy

As shown, the description of the scattered light from a sphere is full described by Eqn.3.7, known as the Lorenz-Mie scattering theory [24]. In the limit of particles on the order of the size of the wavelength of light, the Rayleigh model is an easier, equivalent method derived from Mie theory.

To collect the light, there have been three main schemes used in literature; forward, side, and back-scattering microscopy schemes.

As shown in Fig. 6.7, the scattering distribution taken at different projections widely varies. In addition, the amount of light that gets collected in each of these length-scales also varies.

In the back-scattering scheme, the light collected is the sum of the incident light and the scattered light.

$$I(x,y) = |\vec{E}_r + \vec{E}_s|^2 \tag{3.14}$$

given that the incident wave is known as a polarized plane wave, this equation can be rewritten for the scattered light;

$$I = E_r E_r^* + E_s E_s^* + 2\mathrm{Re}[E_r E_s] \tag{3.15}$$

The incident field will be modulated by a scattering term, which is the Rayleigh scattering amplitude, S, as well as the reference term, R. Thus, the total intensity will be the sum of the scattering amplitude, reference amplitude, and the phase interference between the two fields;

$$I = |E_i|^2 \left[|R|^2 + |S|^2 + 2|R||S| \cos \Delta\varphi\right], \quad \Delta\varphi = \varphi_s \varphi_r \tag{3.16}$$

It should now be noted that the angle of image formation has not been specified. This is of importance as it allows to discuss the various methods of scattering microscopy without loss of generality between methods. Notably, the main difference is the angular content which is obtained from the various scattering models. In the case of spheres, this means using the various angular slices of the resultant scattered field.

3.5 Interferometric Light Scattering (iSCAT)

The key assumption in iSCAT is the relative magnitude of terms. The reference field is constant. The scattering term $R \approx k^4 \alpha^2 = \frac{V^2}{\lambda^4}$. Similarly, the inference term scales as the square root of this. Thus, the scattering term is dominated by the two fields, which we can ignore.

To accurately measure the positions, we can define the contrast metric, which relates the total field to that of a reference field;

$$\begin{aligned} C &= \frac{I - |E_r|^2}{|E_r|^2} \\ &= \frac{|S|^2 + 2|R||S| \cos \Delta\varphi}{|R|^2} \\ &\xrightarrow{|S| \ll |R|} \frac{2|S|}{|R|} \cos \Delta\varphi \end{aligned} \tag{3.17}$$

The phase of the interference consists of a sum of terms consisting of phases accumulated through the traversal of the scattered field. The total phase is dominated by the Gouy phase, which consists of phase accumulated when light is passed through a converging and diverging field, the defocus phase, which can be parameterized to a relative phase shift, as well as the phase accumulated phase difference, which is typical of a Michaelson-Interferomer type system. Experimentally, this system is described in the set of schematics in Fig. 3.1.

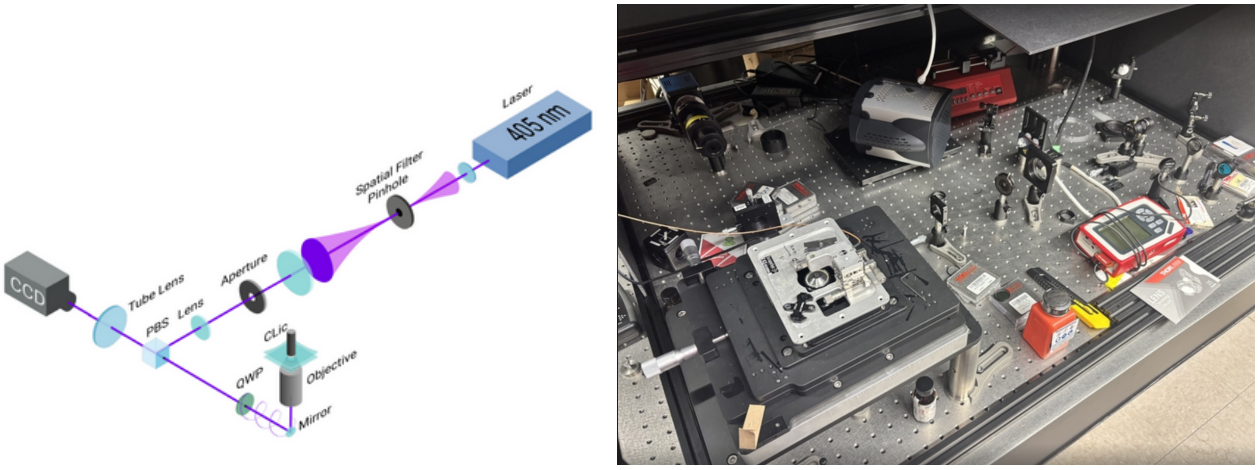


Figure 3.1: Left. iSCAT Schematic. Right. Image of physical system. A 405 nm laser is sent through a spatial filter, passes through an aperture. The beam is sent through a focusing lens which is reflected off a polarizing beam splitter, polarized through a QWP focused on the back plane of the objective. The scattered light is then polarized again, passing through the polarizing beam splitter and is focused on the CMOS camera. The final image is collected via the CMOS and saved in binary format.

3.6 Resolution limits of optical microscopy

The last thing to touch upon is understanding how the image that is being enlarged arrives to the camera, which consists of discussion of optical aberrations.

As mentioned in the formulation for the electric field scatter we have that any optical field consists of a set of plane waves that construct and deconstruct to form a field. That is, the electric field lies in the basis of a fourier series of terms of complex exponential functions. This fact allows one to describe the propagation of a wave in throughout z.

$$\vec{E}(x,y,z) = \iint_{\Omega} \vec{E}_0(k_x,k_y) e^{-i\left(xk_x+yk_y+\sqrt{k^2-k_x^2-k_y^2}\right)} dk_x dk_y, \quad \Omega = \{x \in \mathbb{R} : x^2 \leq k^2\} \tag{3.18}$$

We have that the energy of the wave is fixed by the trigonometric relation of the transverse wave vectors. Notably, this is fixed by the relation $|k| = \frac{2\pi}{\lambda}$.

The propagated wave passes through an objective which can be approximated by a lens. Using the ABCD matrix formulation, we can find that to have a clear image, the imaging condition must be satisfied, that is

$$\frac{1}{z_1} + \frac{1}{z_2} = \frac{1}{f} \tag{3.19}$$

where z_1, z_2 are the positions of the observer and the object, and f is the focal distance of the lens. Thus, the magnification of the lens is defined as $M = z_2/z_1$. However, to understand the aberrations it is necessary to have a field formulation of this problem. The simplest way to do this is described using linear optics theory.

Briefly, we define a pupil function $P(x, y)$ which is unitary if $r \leq R$, where R is the radius of lens. The point spread function of the object is then defined as a delta function through a two lens system. This is given by the fourier transform of the point source masked by the pupil function;

$$H(x, y) = \iint_{\Omega} P(x', y') e^{\frac{ik}{2}\left(\frac{1}{z_1} - \frac{1}{f_1} - \frac{1}{f_2} + \frac{1}{z_2}\right)\sqrt{x'^2 + y'^2}} e^{-\frac{ik}{z_2}(xx' + yy')} dx' dy' \tag{3.20}$$

Expanding this, one can use a zero-order Hankel transform to integrate over the radius and find that the point-spread function, H, is defined in terms of the NA and the magnification;

$$H(r, z) = \int_0^1 e^{-\frac{ik}{2}\frac{\mathrm{NA}^2}{n}\rho^2 z} J_0\left(\frac{k\mathrm{NA}}{M}\rho r\right) \rho d\rho \tag{3.21}$$

where ρ is the rescaled radial vector $\rho = r'/R$, $\mathrm{NA} = nR/f_1$, and $M = nf_2/f_1$.

In perfect focus, we have that $z = 0$, for $z \neq 0$ we have a defocusing aberration which is defined as the secondary exponential. For a point source, which is diffraction limited, we find that the following is the limit of the optical microscope;

$$H(r, 0) = \left|\frac{J_1(k\mathrm{NA}r)}{k\mathrm{NA}r}\right|^2 \tag{3.22}$$

We can solve that the minimum distance between the peak of the PSF and the minimum is solved by $k\mathrm{NA}r = x_0$ where x_0 is the first root of $J_1(x)$. This is classical known as the minimum transverse imaging criterion;

$$r_{\min} = 0.61\frac{\lambda}{\mathrm{NA}} \tag{3.23}$$

Similar results are found for the minimum axial imaging criterion by setting $r = 0$ and integrating.

$$z_{\min} = \frac{2\lambda n}{\mathrm{NA}^2} \tag{3.24}$$

At this point, we invite the fact that not all the light that scatters passes through the lens. Specifically, the range of k_x and k_y are fundamentally limited by the wavelength. The resolution limit is λ/NA. In the case of scattering microscopy, this is not a large problem as the scattering pattern is much larger, however, in the limit of two particle near each other, the spatial resolution limit must be enforced.

3.7 Sources of noise in imaging

In imaging, there is a number of sources of aberrations that one must account for. To discuss these sources of noise, a list of noises will be formed as one goes through the process of taking an image.

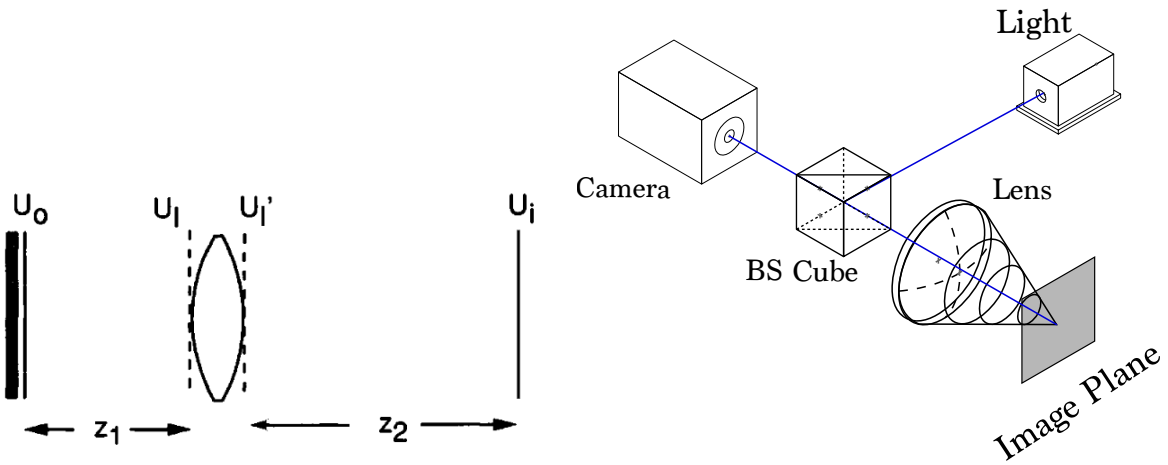


Figure 3.2: Simplest possible imaging setup. Image consists of a plane wave illumination, a lens, and a camera.

The schematic of an iSCAT microscope was already shown in Fig. 3.1, however, one can use a simplified model system without loss of generality. For a general imaging system, one only requires a single lens for imaging. This simple setup is shown in Fig 3.2.

The lens will collects light from the object plane, U_o to the imaging plane, U_i, or the camera's plane. The propagation of that wave will collect aberration from the len's point spread function, as well as phase accumulation. The traversal over the point spread function is

$$U_i(x_i, y_i) = \iint_{\mathbb{R}^2} h(x_i, y_i, x_o, y_o) U_o(x_o, y_o) dx_o dy_o, \tag{3.25}$$

where $h(x_i, y_i, x_o, y_o)$ is known as the optical transfer function, or the response of a lens to a point-source, modelled as a delta-function. The len's magnification is incorporated into this using the following approximation;

$$h(x_i, y_i, x_o, y_o) \approx K\delta(x_i \pm Mx_o, y_i \pm My_o) \tag{3.26}$$

where K is a complex constant and M is a real-valued magnification. We can see that because of this, our image is spatially limited by this point-spread, which was earlier shown by the diffraction limit.

In addition to the spatial noise, there is a phase accumulation known as the Guoy phase. As the light, which can be assumed as consisting of an Eigen-expansion of plane waves, passes through a lens, these plane waves will converge to a gaussian beam with a beam-waist around the focal point. As this focusing occurs, there is a phase that is accumulated known as the Guoy phase. This phase accumulation is equal to

$$\psi(z) = \arctan\left(\frac{z}{z_R}\right), \quad z_r = \frac{\pi w_0^2 n}{\lambda}, \tag{3.27}$$

where w_0 is the waist of the final beam, n is the refractive index, and λ is the wavelength of light. In imaging, as long as the refractive index is constant, there is a total phase shift of π.

In the presence of a background beam, there will be a presence of a background. This background's source is the illumination beam itself, in the case of scattering microscopy. This background noise profile will take the shape of the illumination profile alongside the intensity of the power density.

The camera used will also contain electronic noise associated with it. For CMOS cameras, the noise present will consist of photon noise from the detectors and readout noise from the data-transfer. These noises can be modelled as Poisson noise from the photon detections and a gaussian for the readout noise.

In the case of using an imperfect lens, or a set of lenses as one has with an objective as one does in high magnification microscopy, there is a risk of spherical aberration. Spherical aberrations are caused by imperfections and misalignments in the lens itself. This cases the focal point of the lens to be slightly different depending on which area the light is focused by lens. For imaging purposes, spherical aberrations cause various parts of the image to be out of focus, while others are in focus, creating non-uniform images.

Chapter 4

Biological Application

As we have described, the diffusivity of a spherical particle is directly linked to a motion that is thermally driven. It happens that in the nature of biological cells, this diffusive motion is a driver for many reactions, such as enzyme-protein coupling, protein-protein coupling, and vesicle binding. Thus, it is of utmost importance to adequately understand the confinement effect of diffusion in a given system.

Application-wise, the confinement of nanoparticles is directly related to fabrication and development research. Notably, since nanoparticle fabrication involves methods of quality control, these methods must be able to quantify between different measures of noise impending their measurement. In the case of confinement effects, these effects often hinder diffusion, as described, which can indirectly bias the final reported measurement.

4.1 Diffusion in cells

The cellular system consists of many different components co-existing in the cellular cytoplasm. To get a survey of sizes, the organelles are on the order of $1 - 10\,\mu\text{m}$. while smaller proteins and enzymes, such as ribosomes, vesicles, and lysosomes are on the order of $10 - 100\,\text{nm}$.

This beings said, the motion is directly linked to the fluidic properties of the cytoplasm. In the case of a eukaryotic cell, the Reynold's number is low, which implies an overdamped langevin equation. In addition, the thermal scale is that of high temperature, or $\sim 310\,\text{K}$. In addition, the comparison between diffusion and advection is important. For diffusion to dominate we need that the diffusive timescale is must shorter. Here, we find that since the Peclet number is proportional to the characteristics length L, while the reynolds number is L^2. for small characteristic lengths, such as those proposed, we see that the diffusive term is clearly shorter.

4.2 Brownian motion as a method for measurements of size in biology

As previously shown, we can infer the stoke's einstein diffusion relation to directly measure the size of a biological diffusing particle. In particular, this relation can be explicitly formalized for the hydrodynamic radius of a spherical nanoparticle.

$$a = \frac{k_b T}{6\pi\eta D(\vec{x})} \tag{4.1}$$

Methods such as nanoparticle-tracking analysis (NTA) or single-molecule particle tracking (SPT) are based on this method.

However, there are other methods that can be used. As this thesis is not concerned with fluorescent methods, they will not be explained in depth, however, fluorescence correlation spectroscopy (FCS) works on similar ideas of measuring diffusion autocorrelation curves using a fluorescent probe.

In compliment to single molecule, there are bulk methods as well. In particular, dynamic light scattering (DLS) allows to measure the size of molecules based on their shape function. In particular, the scattering intensity can be weighting to the inverse of a^6, which in turn returns the size of the sample. However, these methods are ill-posed for poly-disperse sample, which is one of the weaknesses of the DLS method.

4.3 Lipid Nanoparticles

The biological molecule of choice for this investigation of confinement is a useful one with lots of applications - the lipid nanoparticle (LNP).

First, it is necessary to identify with the liposome. The liposome, as a macroscopic object, is a spherical membrane of lipid subunit, each carrying a charge and hydrophobicity. The size of a single submit is on the level of angstroms, however, the final liposome can be between $1\,\text{nm} - 10\,\mu\text{m}$. Liposomes and cellular vesicles are very similar, which naturally posed the question of drug delivery using a liposome model.

In recent years, the LNP was formulated - a collection of lipid based walls that are used to encapsulate a drug or cargo. These have applications in drug delivery and have been using in the Sars-Covid vaccine in 2020. These molecules are very useful and thus it is important to know how to improve them.

Modern tools to image and characterize the LNP are split into single-molecule techniques and bulk assays. In the realm of single-molecule methods, CLiC has been shown to be a high-throughput, accurate, and reliable method to characterize these particle's properties.

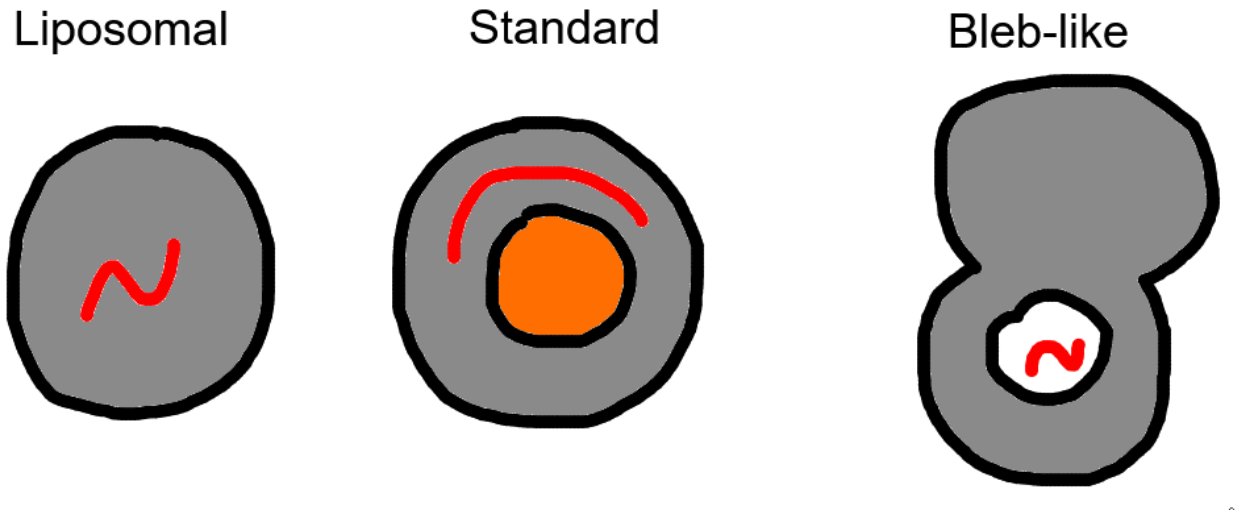


Figure 4.1: mRNA-LNP types diagram. Black is lipids, grey is hydrophobic region, red is mRNA, orange is oil-core.

4.4 LNP topology types

One recent investigation of LNP research is within the functional differences based on the LNP topology type. These topologies have many different types, but typically the topologies are grouped into three main types - liposomal, spherical, and bleb-like. These three topologies not only have different structures, but also different length scales. The topologies are visualized in the figure below.

Chapter 5

Langevin Simulations

To illustrate the wealth of information and form prior distributions for the confined diffusion model, we can simulate the diffusion process of a control molecule diffusing in a confined electrostatic environment.

The goal of this project was to model the diffusion process of confined diffusion using literature, and utilize pre-existing theory to for a statistical estimator of surface charge from diffusion dynamics.

5.1 Simulation Equations

The simulation scheme follows that described in [1]. A brownian particle diffuses between two walls such that (x, y) are perpendicular to the walls, and z is parallel. The electrostatic force that the particle feels is then only in the z direction. The diffusivity of the particle will depend on the distance of the walls. The Langevin equation simulated,

$$m\ddot{\vec{x}}(t) = -\lambda\dot{\vec{x}}(t) + \vec{F}(\vec{x}(t)) + \sqrt{2k_bT\lambda}\vec{W}(t), \tag{5.1}$$

is first separated into transverse and axial components, where F is the electrostatic force, γ is the friction. Here, we can re-write the diffusion coefficient using the Stokes-Einstein relation,

$$D(z) = \frac{k_bT}{\gamma(z)}. \tag{5.2}$$

Then, in over-damped regime, where $m\ddot{\vec{x}} \ll 0$, we can write the equations of motion as

$$\begin{aligned} \frac{dx}{dt} &= \sqrt{2D_x(z)}W_x(t) \\ \frac{dz}{dt} &= \sqrt{2D_z(z)}W_z(t) + \frac{F_z(z)}{\lambda(z)}, \end{aligned} \tag{5.3}$$

as described in Eqn. 7 of [1]. This is exceptionally valid for particles between $10\,\text{nm} - 100\,\mu\text{m}$ as this is exactly the over-damped regime for many particle materials, such as PMMA or coated polystyrene.

To simulate this equation, we use an Euler-Maruyama scheme. To illustrate this, we can rewrite the above equations in the Ito form, replacing lowercase variables with their capital, stochastic, equivalents:

$$\begin{aligned} dX_t &= \sqrt{2D_x(z)}dW_x \\ dZ_t &= \sqrt{2D_z(z)}dW_z + \frac{F_z(z)}{\lambda(z)}dt, \end{aligned} \tag{5.4}$$

The simulated system of equations is then the pair of equations;

$$\begin{aligned} x_{n+1} &= x_n + \sqrt{2D_x(z_n)}W_{x,n}\Delta t \\ z_{n+1} &= z_n + \left[D_z'(z_n) - \beta D_z(z_n)V'(z_n) + \sqrt{2D_z(z_n)}W_{x,n}\right]\Delta t \end{aligned} \tag{5.5}$$

where $V(z)$ is the electrostatic potential, $\beta = 1/k_bT$, and Δt is the discretized time.

5.2 Surface Charge Estimator

As described earlier, the relations for the effective potential are described if one knows the particle properties and surface charge. In terms of measurable electrostatic interactions can be fully described by the following equation,

$$U(z) = B\left[e^{-(H-z)/\lambda_D} - e^{-(H+z)/\lambda_D}\right], \tag{5.6}$$

where the pre-factor is described by the map $f : (\zeta_p, \sigma_w, a, \lambda_D, T)$,

$$B = C \cdot \tanh\left(\frac{e_0\zeta_p}{4k_bT}\right) \cdot \tanh\left(\frac{e_0\psi_w}{4k_bT}\right) \tag{5.7}$$

$$C = \frac{64\pi\varepsilon_0\varepsilon_r a k_b T}{e_0^2} \tag{5.8}$$

$$\psi_w = \frac{2k_bT}{e_0}\text{asinh}\left[\frac{\sigma_w e_0 \lambda_D}{2\varepsilon_0\varepsilon_r k_b T}\right] \tag{5.9}$$

where ζ_p is the zeta-potential of the particle, ψ_w is the stern-layer potential for the surface walls, a is the particle radius, e_0 is the elementary charge value, and λ_D is the Debye-length.

The surface charge of the walls can be estimated if one knows the electrostatic amplitude potential. Since the electrostatic force is a drift term, one can construct a maximum likelihood estimator as a standard result from the SDE. This estimator is derived in App. A.

$$\hat{B} = \frac{\sum_n \frac{\phi(z)}{2D_z}\left[\Delta z - D'(z)\Delta t\right]}{\sum_n \frac{\phi^2(z)\Delta t}{2D_z}} \quad , \mathbb{V}[\hat{B}] = \sum_{n=0}^{\infty} \frac{\phi^2(z)\Delta t}{2D(z)}. \tag{5.10}$$

If the amplitude of the electrostatic interactions is known, one can invert the amplitude using relations in Eqn. 5.7. This inversion requires at least the zeta-potential of the particle, the size, and Debye-length.

5.3 Results

We will simulate the priors of a Polystyrene bead diffusing with neutral charged walls, as well as those with varied levels of screening. In turn, we will be able to showcase the different cases of the known confined diffusion theory, as well as establishing prior curves for this work. In addition, the simulated curves will be compared to an effective theoretical model. For a given simulation, we can generate a realization of the dynamics following the aforementioned SDEs. These realizations approach the Boltzmann limit as $t \gg \tau$. Some such realizations are shown in Fig. 5.1

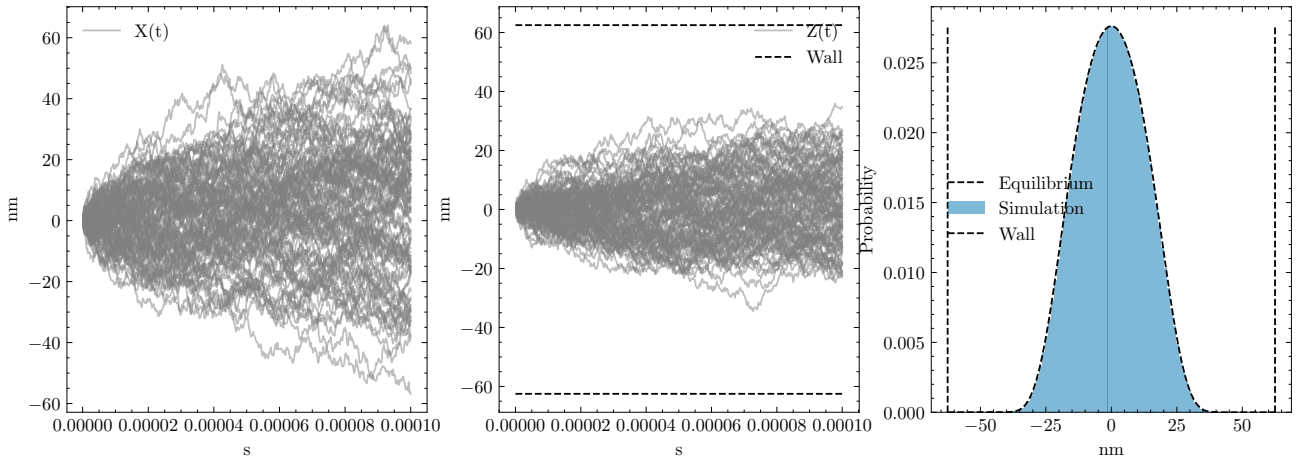


Figure 5.1: Confined Brownian Particle realization of z diffusion in electrostatic potential. $a = 50\,\mathrm{nm}, \zeta_p = -30\,\mathrm{mV}, \sigma_w = -2\,\mathrm{mC/m^2}, L = 225\,\mathrm{nm}, \lambda_D = 10\,\mathrm{nm}, T = 300\,\mathrm{K}, \eta = 1\,\mathrm{mPa\,s}$. Left. Particle trajectory vs time for 100 realizations. Right. Histogram of steps vs. Boltzmann distribution.

5.3.1 Mean-Squared Displacement in Confinement

For a given experiment, one aims to measure something from knowing the trajectory of the particle. The simplest is an estimate of the size of the particle. One can do this using the transverse and axial trajectories obtained from the Langevin equation as a method to test the variance of the estimator given a known ground truth. In the case of an electrostatic potential, we can directly how measurements which do not account for screening effects change the measurement.

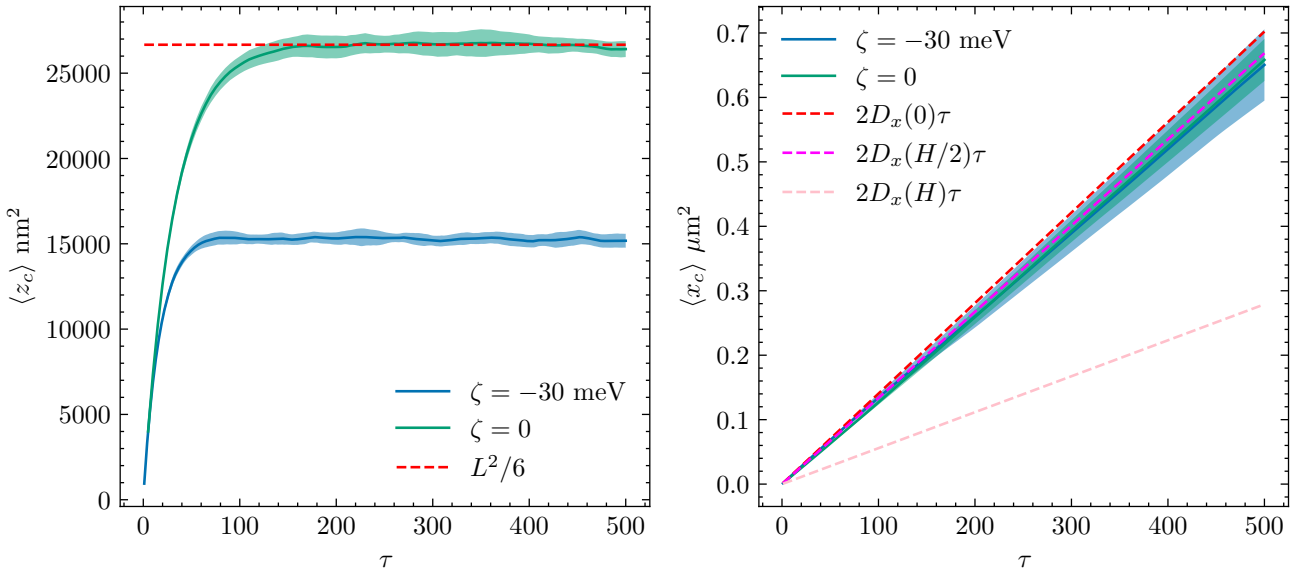


Figure 5.2: Comparison between MSd curves of two highly different confined diffusors. Exposure time $\tau = 200\,\mu s$, $\lambda_D = 10\,nm$.

The notation of a higher confinement with respect to the zeta-potential is intrinsically linked to the electrostatic potential that the particle feels as it explores the well. This relationship between the effective confinement length can be solved by solving the Green's function for the FP-equation in the presence of an electrostatic potential. However, this happens to be non-analytical in the case of the Poisson-Boltzmann potential approximated in the Debye-Huckle theory. However, using the developed simulation class, this relationship can be directly measured. The same settings were used while fixing the zeta-potential of the particle, and varying the ionic concentration of the solution, which is inversely proportional to the Debye-length,

$$\lambda_D \propto \frac{1}{\sqrt{C}}, \tag{5.11}$$

where C is the ionic concentration of a dilute electrolyte compound [CITE]. To model this, the simulation settings are chosen as such $a = 50\,nm, L = 500\,nm, T = 300\,K, \Delta t_i = 0.2\,\mu s, \tau = 200\,\mu s, \eta = 1\,mPa\,s$ and the Debye length is varied. The zeta potential of the particle and glass are chosen to resemble glass and a polystyrene bead, fixed at $\sigma_w = -2\,mC/m^2, \zeta = -30\,mV$. The relationship between the effective confinement length and the ionic concentration is shown in Fig. 5.3

The relationship between the effective confinement length, as well as the relationship between relative diffusion and concentration of ions begs the question of if this can be used as a particle filter to distinguish classes of particles. Here we investigate that question regarding the amount of confinement length compared to the zeta-potential of a particle at various Debye-lengths. This relationship is described in Fig. 5.5.

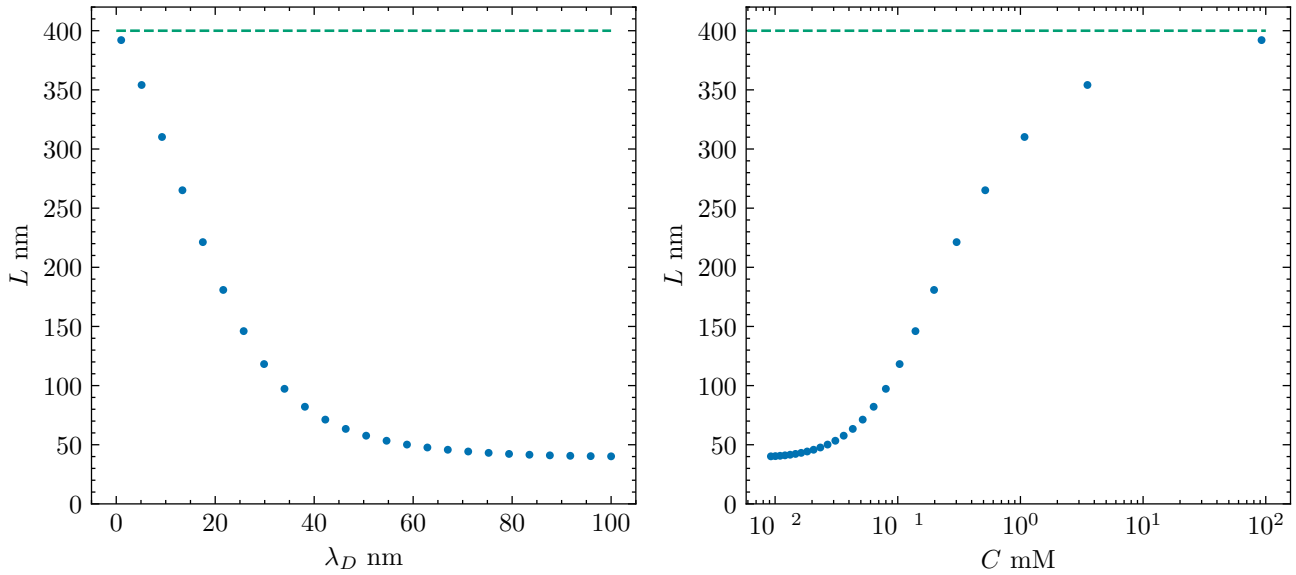


Figure 5.3: Effective Confinement length plotted against the solution's Debye length. $a = 50\,\mathrm{nm}, L = 500\,\mathrm{nm}, T = 300\,\mathrm{K}, \Delta t_i = 0.2\,\mu\mathrm{s}, \tau = 200\,\mu\mathrm{s}, \eta = 1\,\mathrm{mPa\,s}$ and the Debye length is varied. The zeta potential of the particle and glass are chosen to resemble glass and a polystyrene bead, fixed at $\sigma_w = -2\,\mathrm{mC/m^2}, \zeta = -30\,\mathrm{mV}$.

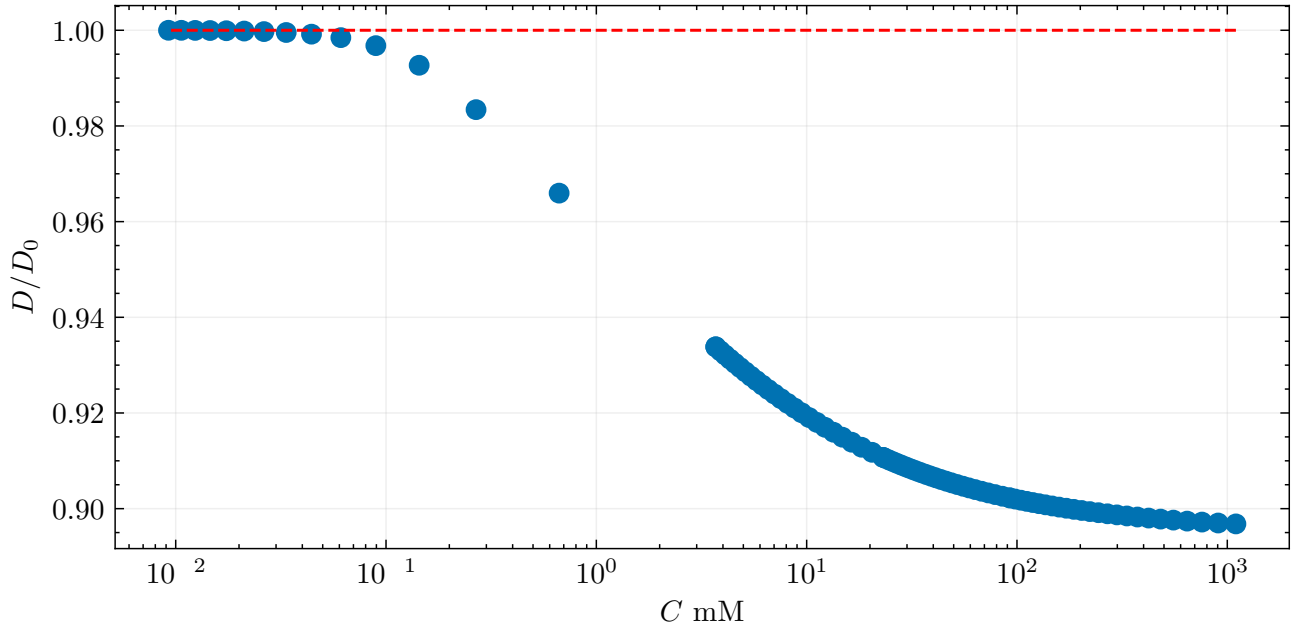


Figure 5.4: Effective diffusion plotted against the solution's Debye length. $a = 50\,\mathrm{nm}, L = 500\,\mathrm{nm}, T = 300\,\mathrm{K}, \Delta t_i = 0.2\,\mu\mathrm{s}, \tau = 200\,\mu\mathrm{s}, \eta = 1\,\mathrm{mPa\,s}$ and the Debye length is varied. The zeta potential of the particle and glass are chosen to resemble glass and a polystyrene bead, fixed at $\sigma_w = -2\,\mathrm{mC/m^2}, \zeta = -30\,\mathrm{mV}$.

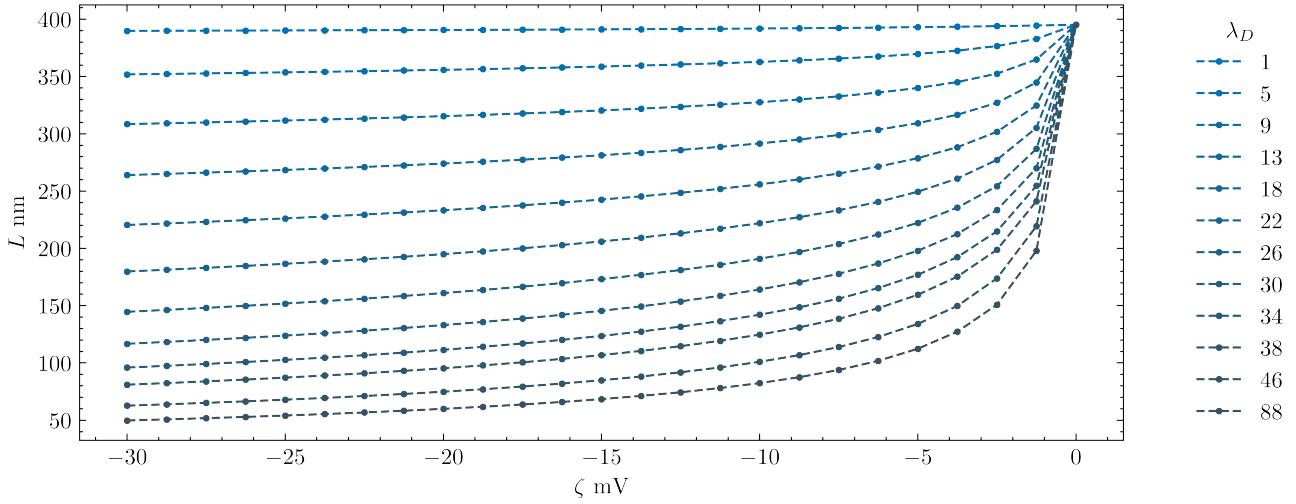


Figure 5.5: Effective confinement plotted against the particle's zeta-potential. $a = 50\,\text{nm}, L = 500\,\text{nm}, T = 300\,\text{K}, \Delta t_i = 0.2\,\mu\text{s}, \tau = 200\,\mu\text{s}, \eta = 1\,\text{mPa}\,\text{s}$ and the Debye length is varied. The surface charge of the wall are chosen to resemble glass, fixed at $\sigma_w = -2\,\text{mC}/\text{m}^2$.

5.3.2 Surface Charge estimation

The particle's exploration demonstrated a clear dependence on a drift due to the electrostatics. This drift is a useful estimator of the particle-wall charge relationship. Using the aforementioned amplitude maximum-likelihood estimator, drift estimator can be extracted into meaningful, physical information. The Langevin simulation was ran for 20 ms with time-step of 0.2 µs. The distribution of the MLE is shown in Fig 5.6.

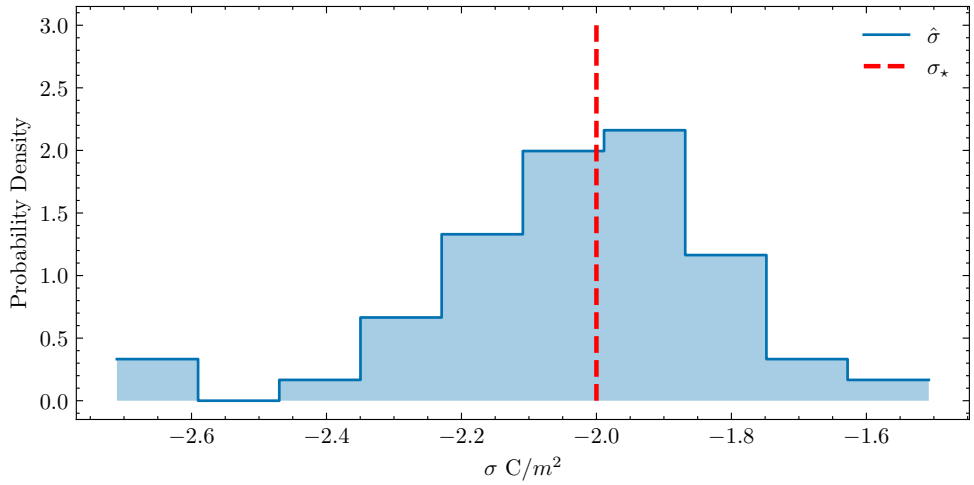


Figure 5.6: Distribution of $\hat{B}$ Re-parameterized for the surface charge. $\hat{\sigma} = 1/(A \cdot C)^2 \hat{B}$ where A, C are properly defined in 5.7. In this histogram there are 50 simulations with seeds set to $S_{\text{eed}} = i * 10 + 1$ for $i \in \{0, 49\}$ using the *numpy* seed parameter. $a = 50\,\text{nm}, L = 500\,\text{nm}, T = 300\,\text{K}, \Delta t_i = 0.2\,\mu\text{s}, \tau = 200\,\mu\text{s}, \eta = 1\,\text{mPa}\,\text{s}, \lambda_D = 10\,\text{nm}$ The surface charge is fixed at $\sigma_w = -2\,\text{mC}/\text{m}^2$.

5.4 Conclusion

In conclusion, we have shown that confined simulations can be readily simulated with physical parameters as per literature. The simulations have shown that there are regions of instability in the Debye-surface charge regime where the confined diffusivity of the particles is maximally observed. For the case of polystyrene beads, this appears to be around $\lambda_D \approx 80\,\text{nm}$.

With regards of surface charge estimation, we have introduced a new MLE for the amplitude based on maximum likelihood estimation. The distribution has a mean centered around the ground truth with a standard deviation of $1/4 \cdot \sigma_\star$.

Chapter 6

iSCAT z-estimation Simulations

This section presents a computational method for determining the axial position of a nanoparticle from iSCAT imaging data.The results obtained are used to evaluate the accuracy and performance of the proposed method for determining the nanoparticle's z-axis position. The precision and limitations are discussed in the context of recovering three-dimensional particle trajectories.

The raw images are acquired from a CMOS camera as per the optical schematic shown in Fig. 3.1 and Fig. 7.1. The images consist of 2D interference images acquired at some z_{focus}, which is the distance between the objective's focal distance and the objects in the field of view.

6.1 Method Overview

The method described in this section is based on computational image processing on images acquired using a CMOS camera. The camera has an array of 512×512 pixels. The image-processing workflow used to determine three-dimensional trajectory is shown in 6.1. The diagram summarized the sequence of image processing steps to extract each coordinate position.

1. Raw image captured on camera.
2. Background subtraction to convert to iSCAT image.
3. iSCAT (x, y) Tracking
4. Radial Integral to obtain scattering data.
5. Nuisance parameter estimation to ensure physical stability.
6. Particle axial position Linking with 2D chain-link algorithm.
7. Axial position Extraction
8. Three-dimensional trajectory is recovered.

Figure 6.1: Background subtraction outline.

6.2 Raw image background subtraction

To obtain high-contrast interference signal from raw camera data, a post-processing background subtraction routine is introduced. This background subtraction measured a temporally averaged background using a window-filter and Fourier filtering. An example of this process is shown in Fig.6.2.

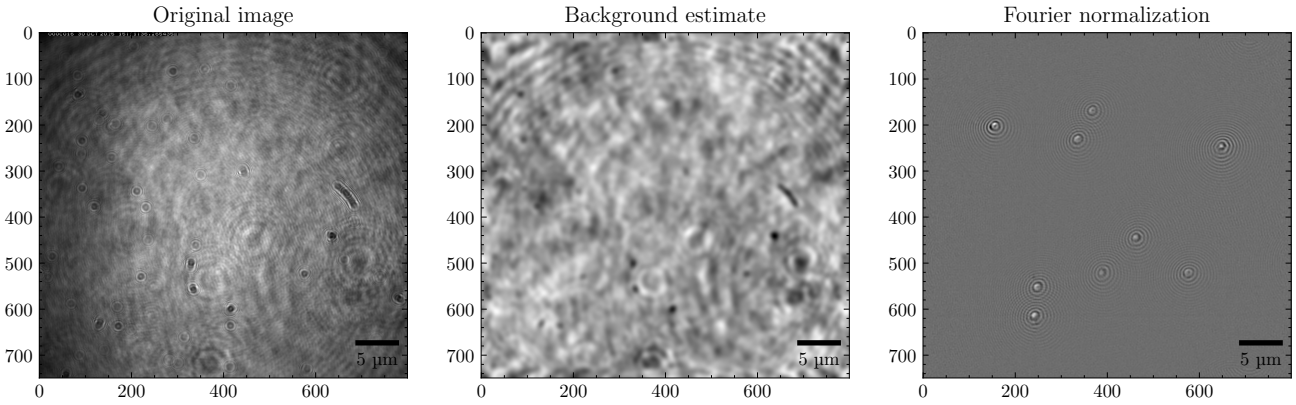


Figure 6.2: Raw image captured on camera to background subtracted iSCAT image. Single particles are tracked and extracted for further analysis.

The full algorithm is presented in Alg. 1. The overview of the algorithm is to subtract the background noise in order to minimize Poisson, and electronic noise.

Poisson noise subtraction is performed by subtracting off the mean background fluctuations. A Poisson background has a constant mean, which is reduced when subtracted away.

In order to remove electronic noise from data-transfer, a row-wise averaging scheme is implemented. Row-wise means are taken and normalized with respect to their row as the data-transfer from CMOS to computer is by horizontal lines. Thus, to ensure all lines are equally weighted, they are normalized with respect to their entire row. This is implemented as per prior iSCAT normalization methods by the Sandoghar lab Vahid's github should be here.

The set of images, which consist of matrices of pixel-values of shape $F \times H \times W$, or F number of frames, W pixels wide, H pixels tall, are passed as entries into the function.

Before running the function, a dynamically allocated setting for the Fourier-filtering is determined based on the image dimensions. Specifically, a donuts of radii inner 1% and outer 25%, roughly translating to $\max(H, W)/100 - \max(H, W)/4$.

The edge 20 pixels of the image stack are ignored using a mask filter. On the inner pixels, the mask is set to 1, while on the pixels less than 20 pixels away from the edge are set to 0.

For each frame in the stack, the image is first normalized using a row-wise normalization function. The function, titled AmpNorm in Alg. 1, first obtains the masked-image. The global normalized

mean is stored. For each row, the mean is taken and the row of the global-mean is adjusted by the row-wise mean. The final image is the global mean divided by the row-mean for each row.

Once normalized, the background of the image is calculated by taking the forward and backward ΔL frames spaced by a spacer of Δf frames and taking the AmpNorm of these frames.

The iSCAT image then consists of a background subtraction between the normalized image and the background. Finally, once background subtracted, the image is Fourier-filtered using the pre-allocated band-pass filter. This process is demonstrated in Fig. 6.2

6.3 iSCAT (x, y) Tracking

To determine the transverse positions of any particles detected in the iSCAT method, the radial variance transform algorithm is used [26]. The original algorithm is briefly outlined below for completeness. The full code is written by the Sandoghdar group and can be found in Ref [26]. A demonstration of this method is presented in the Fig. 6.3.

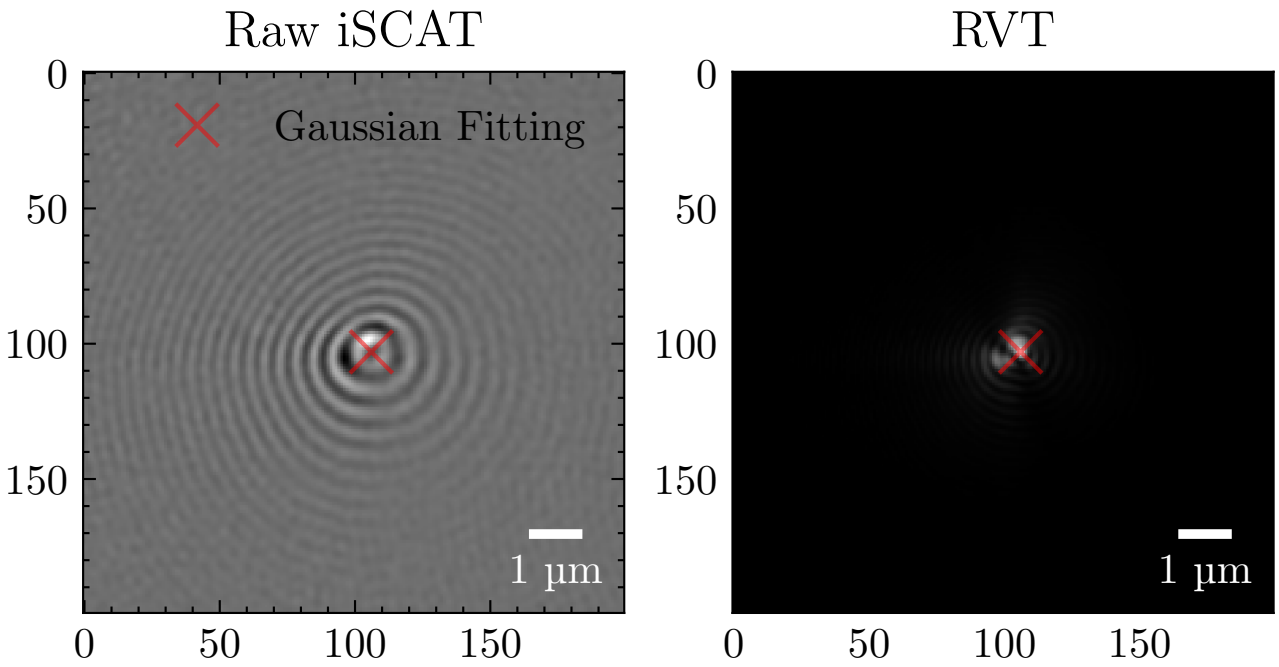


Figure 6.3: Raw image to RVT. Gaussian fitting of the RVT is shown on both plots.

The transverse localization algorithm, referenced as TransverseLocalization in 2 consists of transformations onto the iSCAT image from the previous section.

The input image stack is first loaded and assumed to consist of radially symmetric interference fringes. The image is first transformed using the Radial Variance Transform [26]. This function converts a radial-fringe pattern into a point-like delta function when a radial symmetry is acted upon it.

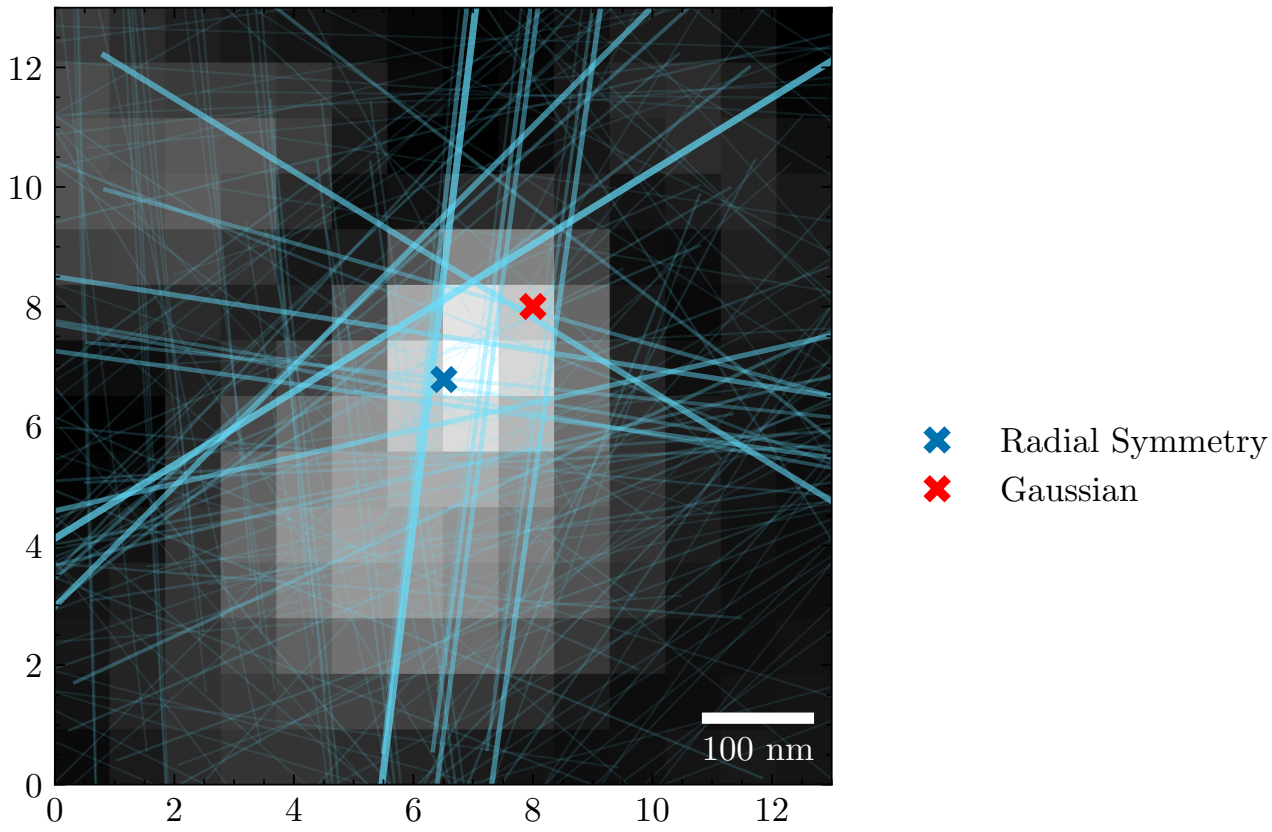


Figure 6.4: Demonstration of the Radial Symmetry method applied onto RVT of iSCAT data. Implemented from [27]. Figure adapted from [27].

Before proceeding, a frame-independent parameter for threshold SNR amplitude, number of pixels, and ROI cropping parameters are inputted. for a particle to be detected, the SNR of the RVT should be roughly 90% of the max iSCAT amplitude. An estimate of particle number is assumed, which acts as a prior for top n detections. Lastly, ROI for sub-pixel estimation is chosen.

For each frame, the iSCAT image is dilated. Particle positions that are considered maxima if the dilated image is equal to the RVT and the RVT is above the threshold for detection.

Once candidates are chosen, a NL-LS gaussian fitting to the RVT is performed to acquire higher-accuracy localization. To improve this to sub-pixel, a crop is done around each RVT gaussian fitted point and a radial-symmetry algorithm is performed as per [27].

To ensure accurate linking, an implementation of the MUNKRES algorithm is used REF. Briefly, a cost-matrix of distances is constructed and minimized in time to ensure that optimal displacements are chosen. This algorithm has limitations in cases where particles overlap. Ideally, experimental measure are taken to minimize this occurring. One such measure is to lower the concentration of particle in the recorded images.

6.4 Angular Integral

Image data will have background misalignment due to iSCAT hardware setup. To optimize for fitting speed and misalignment property, radial scattering profile are assumed as in the Rayleigh regime model is symmetric. The radial integration method is described in Alg 3.

The discrete angular integral consists of summing angular terms on a radial grid. Mathematically, the output array obtained is described by the relation

$$g_t(\rho) \tilde{-} \frac{1}{\Delta\theta} \int_{\theta_0}^{\theta_0+\Delta\theta} I_t(\rho,\theta) d\theta, \tag{6.1}$$

where θ_0 is a constant, $\Delta\theta$ is the width of integration, $I_t(\rho,\theta)$ is the iSCAT image.

First, the cartesian image is Re-parameterized into polar coordinates using a change of basis transformation on the index grid of the image. Then, for each ring of indices, the image coordinates are summed up on this ring. The final array is an angular average that is indexed at radius r for some upper radial bound. This process is illustrated in Fig. 6.5.

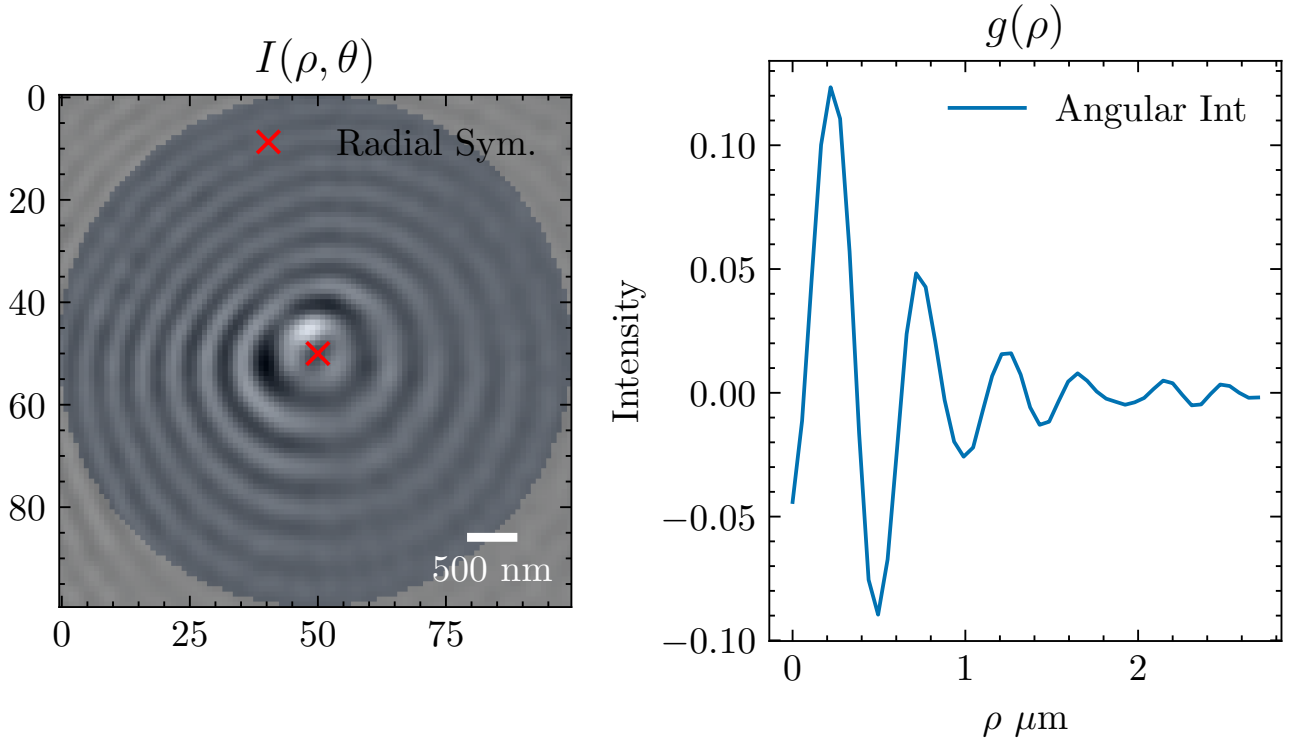


Figure 6.5: Angular Integral demonstration.

6.5 Nuisance parameter Estimation

In order to speed up computation and produce a more statistically relevant result, the method aims to fix the model parameters which should theoretically stay constant. These, for example,

are the particle mass signal and optical aberration coefficients. These parameters, known as nuisance parameters, should be fixed before fitting to the more sensitive parameter of interest, the relative z position.

To obtain accurate estimation of the z position relative to nearby frames, nuisance estimation is done to focus on z fitting. To do this, L-BFGS-B Fitting is performed. alongside amplitude fitting. The theory of nuisance parameter estimation is described in [28].

The nuisance parameter method works on the premise of a model function. The iPSF model described in 3 is used as a weight function to compare to the data. First, bounds are placed for physical values of each parameter.

Here, it is important to note that the amplitude of the signal is proportional to the mass. The spatial decay of the aberration is modeled as a decaying exponential.

For each candidate decay parameter, the weighted correlation function is calculated. Let g be the radial profile, and G the theoretical iPSF model. The inner product of these two arrays, weighted in some integral of radiuses, is defined by $\langle G, g\rangle_w$. The correlation function is then written as

$$\rho(z,\gamma) = \frac{\langle G, g\rangle_w}{\sqrt{\langle G, G\rangle_w + \langle g, g\rangle_w + \varepsilon}}, \tag{6.2}$$

where ε is a small number to prevent division by 0.

The best-fit amplitude function here is analytical, given by the expression

$$\hat{E_0} = \frac{\langle G, g\rangle_w}{\langle G, G\rangle_w}. \tag{6.3}$$

The loss-function, $\mathcal{L}$, given the correlation, is given by the subtraction of ρ^2,

$$\mathcal{L} = 1 - \rho^2(z,\gamma). \tag{6.4}$$

Once a fitted $\gamma_0, \hat{E_0}$ are chosen, the optimal z for each frame is given by a non-linear bounded least squares which are acquired using the L-BFGS-B method with a Huber Loss ($\delta = 1/2$).

When computed for each frame in the stack, one has F optimally chosen γ_0, E_0. To obtain the optimal parameter, a fit to distributions, inferred by [2], is used. These fits include a Gamma distribution for E_0 and a LogNorm distribution for γ. Together, these form nuisance estimation for E_0, γ_0.

6.6 2D Chain-link

Given an estimation of the nuisance parameters, a grid-search of the optimal z is simple. The physically best z position is a mean-squared error loss minima of the model subtracted with the data. However, due to the periodic nature of the data, there are up to 3 candidates for the optimal z position based on $\lambda/4$ having a high degree of similarity. Thus, to resolve this, a z-linking method was implemented to find the most likely trace based on diffusion probabilities.

To link consequent positions in space, a dynamic programming algorithm is used. The algorithm is known as the Viterbi method and is described in Alg. 5.

First, all minima for every frame must be collected. These are the candidate positions that trajectories are constructed from. The losses of these positions are also stored to ensure no spurious candidates are chosen.

For each position, the forward pass compares all chains moving in time. The loss is accumulated by a weighted Euclidean transition cost,

$$d_i(i,j) = \sqrt{w_x^2(x_{t-1}^{(}i) - x_t^{(}j))^2 + w_y^2(y_{t-1}^{(}i) - y_t^{(}j))^2}, \tag{6.5}$$

where x denote z motion, and y denotes the loss difference. This matrix compares all weighted transitions. The optimal transition $\pi_t(j)$ is then chosen to minimize this transition, and any prior transitions.

For each following forward pass, the cost is accumulated with a memory of 2 frames. The memory is chosen to ensure that an incorrect transition in a given frame is not penalized later on.

Thus linking method is thus a search algorithm for the shortest math based on the weighted Euclidean method. A short toy-model is shown in Fig. 6.6 to visualize the procedure.

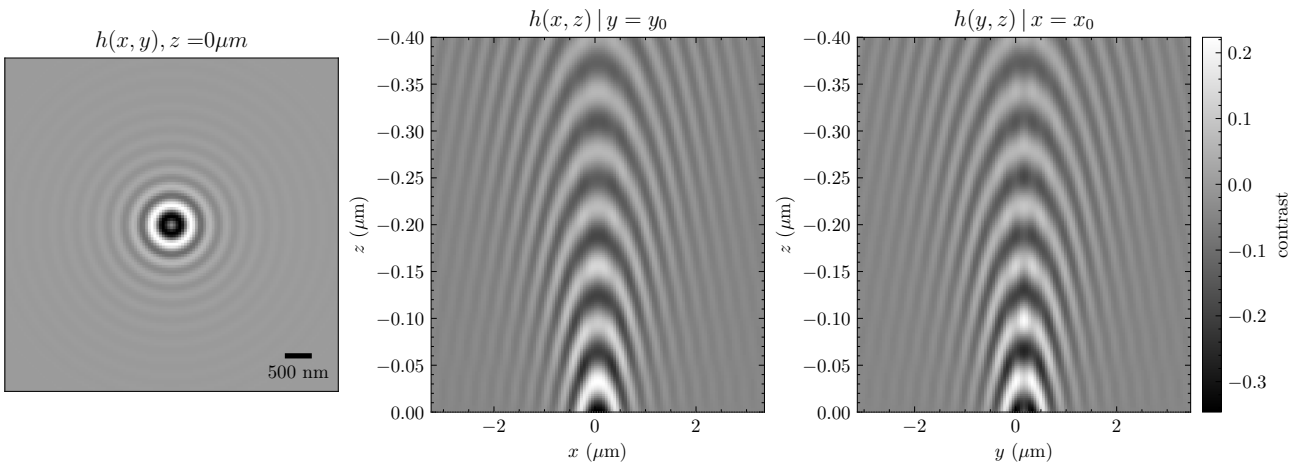


Figure 6.7: Simulated iPSF. Left. Projection of in-focus plane. Middle. x, z projection. Right. y, z projection. Focal scan is over 400 nm. pixel size is 55 nm/pixel. $\lambda = 405$ nm. Assumed particle radius $a = 50$ nm.

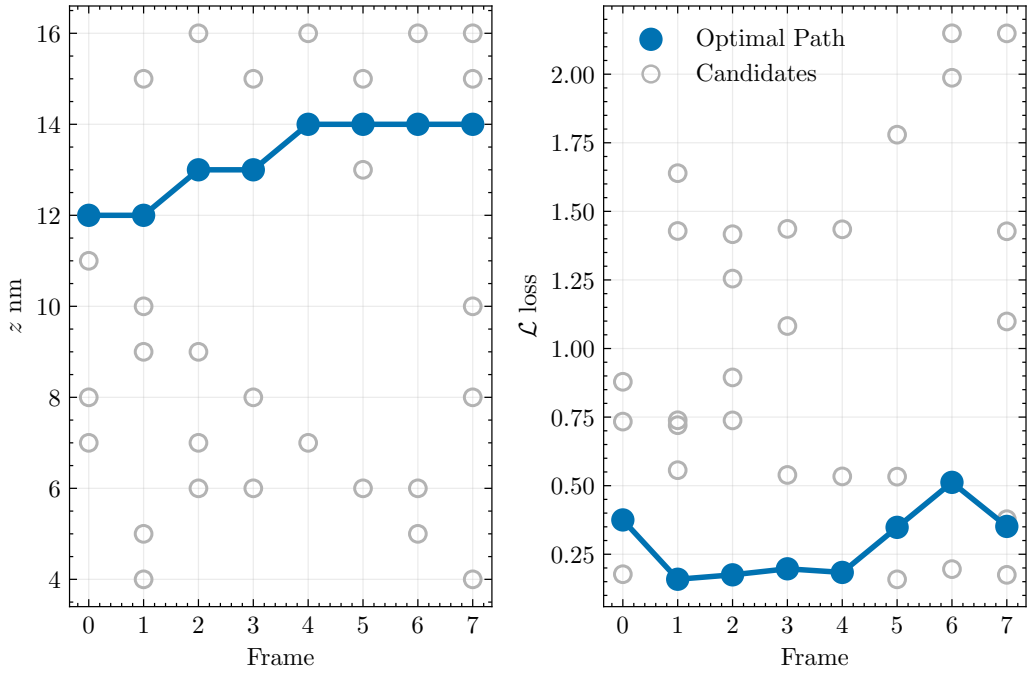


Figure 6.6: Viterbi Search algorithm toy example. Optimal path is chosen based on z similarity and lowest loss.

6.7 Simulated iPSF localization

To validate the proposed method, one performs simulated axial-estimation. Simulated iSCAT point spread functions were generated using the PISCAT library for $\lambda = 405$ nm. The iPSF was generated for the focal range $z_f \in [-1000, 0]$. The assumed particle is spherical with radius $a = 50$ nm. The iPSF is shown in Fig. 6.7

As mentioned, the localization pipeline utilizes a weighting scheme to account for objective

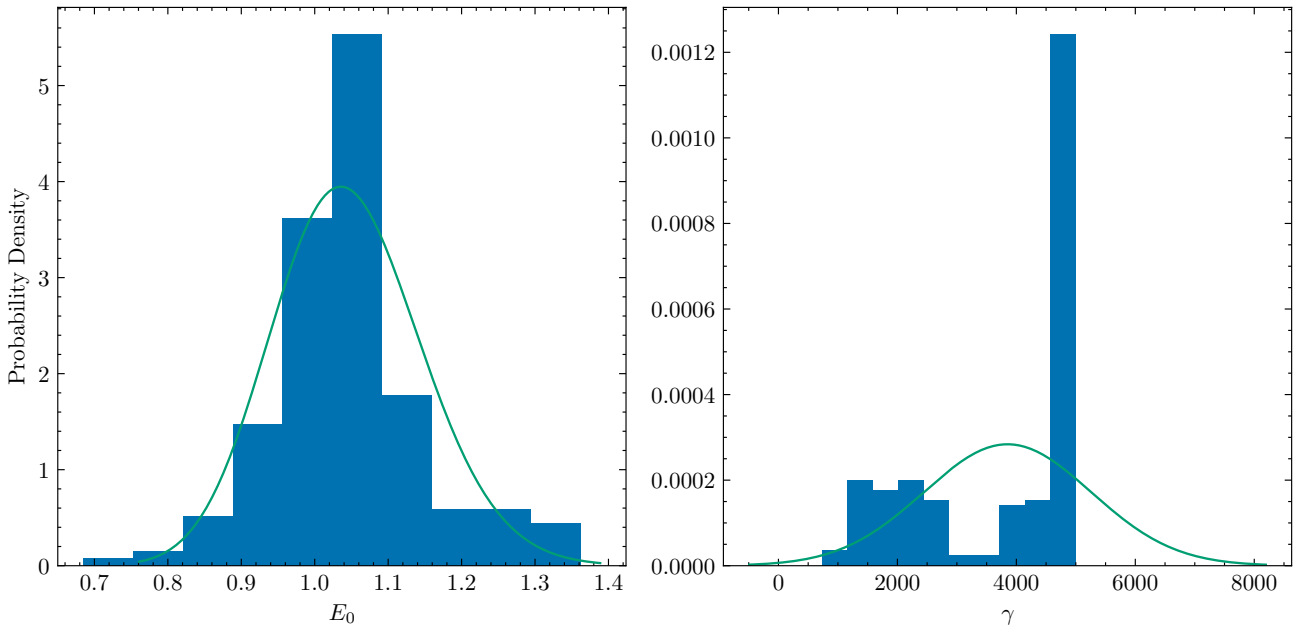


Figure 6.8: Nuisance Parameter estimation for iPSF fits. Left. Amplitude factor. right. exponential decay factor. The exponential decay is not of utmost importance as it often is very small. it is kept for stability purposes.

aberration and misalignments to non-modeled regions of the data, such as the central fringe. This is accounted for by using a binary weighting array, $\mathbf{w}$. In the case of these idealized simulations, we will show the effect of changing the weighting scheme as we proceed. This demonstrates a loss of absolute precision, however, maintains the relative precision, which is that of interest in measurements of axial trajectory.

6.8 3-parameter Estimation

As a reminder, the model for the iPSF is a multivariate function iPSF : $(z, E_0, \gamma) \rightarrow \mathbb{R}^2$. It has been shown that the statistical model of iPSF takes;

$$\begin{aligned} \hat{E_0} &\sim \text{Gamma}(\mu_{E_0}, \sigma_{E_0}) \\ \hat{\gamma} &\sim \text{logNorm}(\mu_\gamma, \sigma_\gamma) \end{aligned} \tag{6.6}$$

The so-called 3-parameter estimation is a sampling of nuisance parameters in the model. The nuisance parameters here are $\hat{E_0}, \hat{\gamma}$, where $\hat{E_0}$ is an amplitude factor, and γ is an exponential decay factor used to account for spatial decay in signal. The amplitude factor can be analytically solved for using a grid-search sweep along z. The γ factor is fitted to using the L-BFGS-B minimization algorithm. This nuisance parameter fit is shown in the histograms of these two parameters shown in Fig. 6.8.

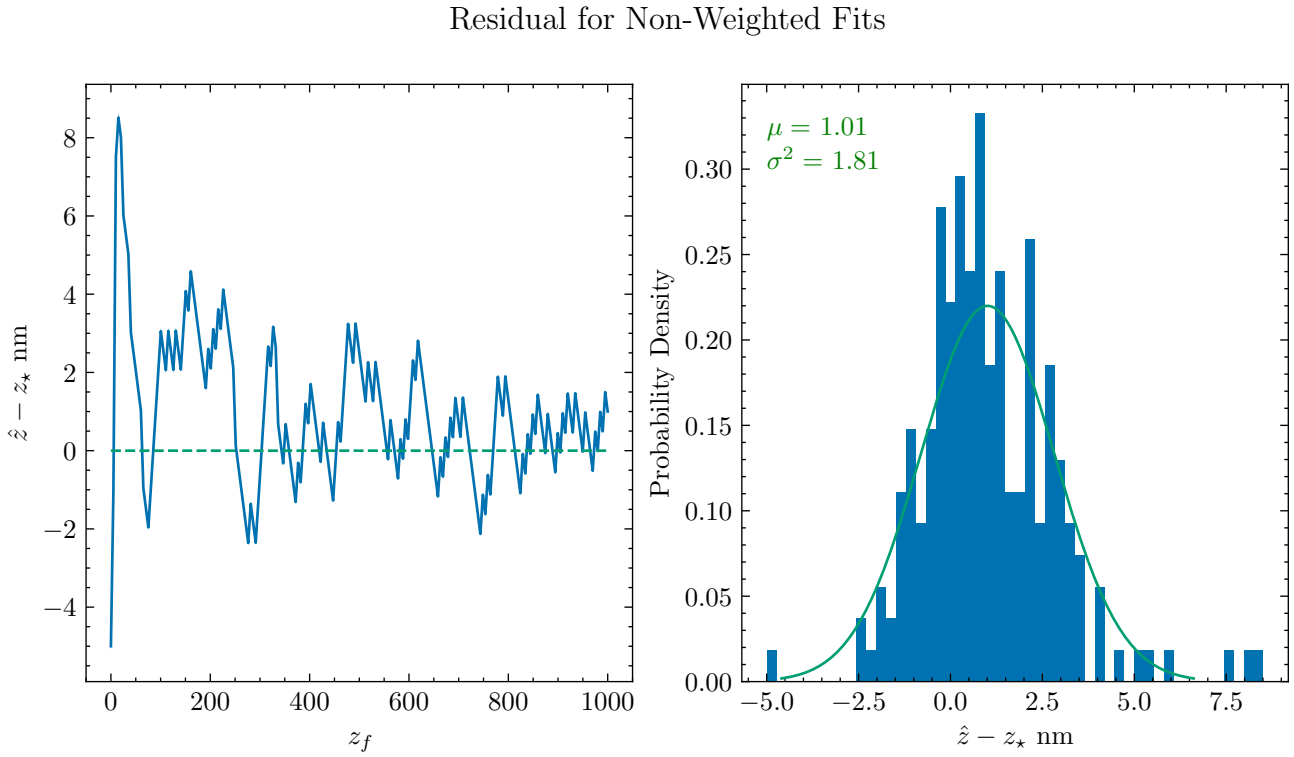


Figure 6.9: Fit residuals using no weighting scheme. reporting an absolute bias of 1 nm with variance of 2 nanometers.

6.8.1 Grid-Search Estimation

Once the nuisance parameters are chosen, a grid-search with a dynamic linking is performed. Each frame is assumed to have a maximal diffusion time. This is in place to optimize for noise and have coherent, physical traces without outliers. The grid search takes the difference between the data and a scan over the model with the nuisance parameters in place. The best loss is chosen. In the case of linking, the minimal loss over the path is chosen. The full algorithm is discussed elsewhere. the residuals of this using no fitting are shown in Fig. 6.9.

6.9 Conclusions

In conclusion, the pseudocode for the method was outlined explicitly. The methodology was tested on simulation data with a high degree of accuracy. The method shows an accuracy of ± 1 nm with a forward bias of 1 nm.

Chapter 7

3D confinement with iSCAT

The goal of this work is to investigate and measure the size-diffusivity relationship of single-molecules in a confined environment. The goal of this project was to better understand the biases, variance, and details of size measurements performed by convex-lens induced confinement (CLiC).

This study overviews a the dynamics on a model particle, which was chosen as plastic beads. The ionic concentration, as discussed in Chp. [5], is very important in focusing particles into an effective confinement length. This paper tends to evaluate this model experimentally, as the Langevin model, although complete, is not-trivial in fuzzy/applied systems. To show this, the method is applied to 3 species of lipid-nanoparticle sample - liposomal, spherical, and bleb-like topologies.

During experimental trails, it has been found that these samples tend to stick to negatively charged surfaces, such as negatively charged cover glass, due to Van der Walls forces and electrostatic interactions. Over the last decade, passivization methods were used to alleviate this sticking. This paper aims to measure these interactions to determine the biases on size characterization associated with measuring molecule size in systems with these passivization methods.

7.1 Methods

The following section outlines the tools used to collect data for this thesis.

7.1.1 Aparatus schematic

The schematic of the system is Here we re-draw Fig 3.1 to remind the reader of the system. The optical system used for imaging is an iSCAT microscope.

The optical system consisted of a 405 nm excitation laser. The output of the laser was 240 mW

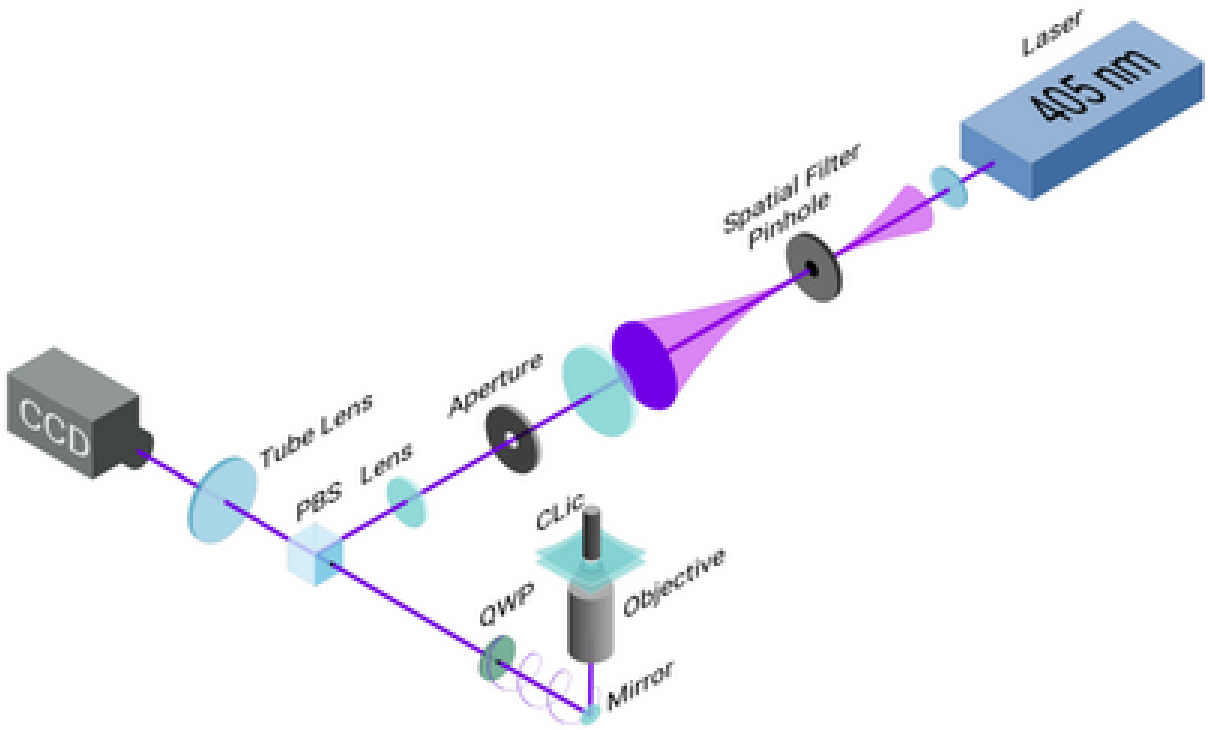


Figure 7.1: iSCAT Schematic. A 405 nm laser is sent through a spatial filter, passes through an aperture. The beam is sent through a focusing lens which is reflected off a polarizing beam splitter, polarized through a QWP focused on the back plane of the objective. The scattered light is then polarized again, passing through the polarizing beam splitter and is focused on the CMOS camera. The final image is collected via the CMOS and saved in binary format.

pre-spatial filter, with an output of 120 mW post-spatial filter. The objective used was the 100× oil imaging objective (Nikon, Apochromat TIRF 100× oil immersion objective N.A. 1.49, W.D. 0.12 mm, F.O.V 22 mm). The camera used was the (PRODUCT ID) CMOS camera. Operating software was custom written by (Author) in LabView.

7.1.2 Cover-Glass

The flow-cell device consisted of two glass substrates (Ted Pella, product no.260372, 145 ± 15 µm thick, 25×25 mm cover glass), separated by a 30 um thick double-sided adhesive spacer (Nitto Denko, product no. 5603).

7.1.3 Convex-lens induced confinement

The CLiC system was used to achieve nanometer confinement while imaging. This method is described in detail in the seminal publication [29]. A convex pusher lens is used together with a piezo linear actuator to displace the top coverslip of the flow-cell in a parabolic displacement. The piezo actuator is manually controlled during imaging and set to a specific, controlled height.

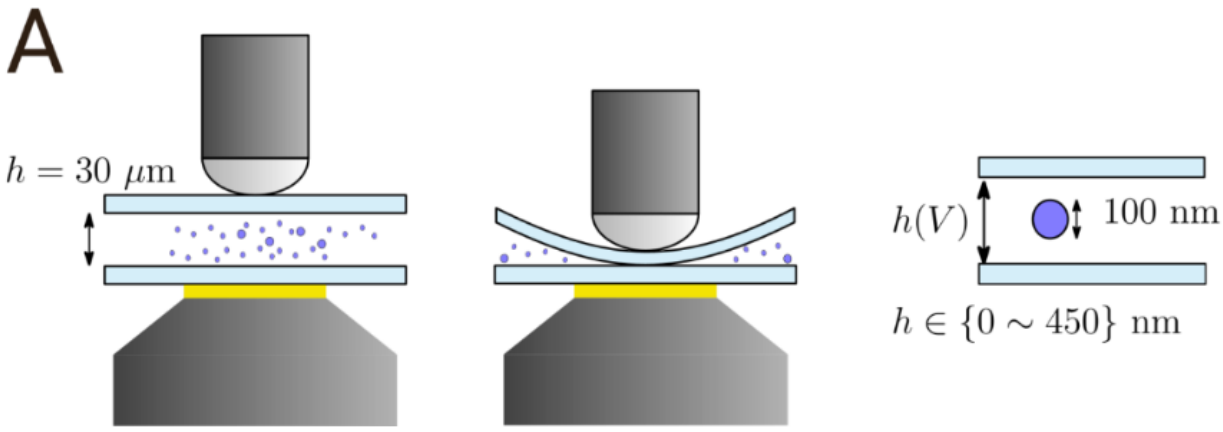


Figure 7.2: Figure Caption.

The height is determined using the inference fringes of Newton's rings. The full method outline is visualized in Fig. 7.2

7.1.4 Sample Details

The samples used in this thesis are listed below. Carboxylate modified polystyrene beads were purchased from Thermofisher (FluoSpheres™carboxylate-modified microspheres, Catalog No. F8799, are 0.1 µm polystyrene beads). Sample consists of 0.1 µm polystyrene beads with 500 µm fluorescent tag attached. The tag is attached as that is what the lab had and has used in prior measurements. The beads were able to be measured in de-ionized water (MilliQ), as well as a concentration series of phosphate-buffered saline.

Lipid-nanoparticles used in this thesis were formulated by the Pieter Cullis lab. The liposomal formulation was used as liposomal (20% norMC3). The spherical, oil-core LNP, is the Onpattro (50% norMC3) LNP formulation made with 25 mM NaOAc. The bleb-like LNP is the Onpattro bleb (50% norMC3) LNP formulation made with 300 mM NaOAc. (Cryo (?)).

7.1.5 Analysis Method

The analysis method is described in detail in prior chapters. The visual pipeline of the method includes the main inputs and outputs. The raw image is converted into a trajectory, which includes sizing information from the MSD. This visual graphic is shown in Fig. 7.3.

7.2 Results

This section describes the results obtained from the experiments. First, the measurements of ionic concentration on polystyrene is presented by comparing the sizes recovered from transverse and axial diffusion, as well as the positional histograms. Second, measurements of axial diffusivity

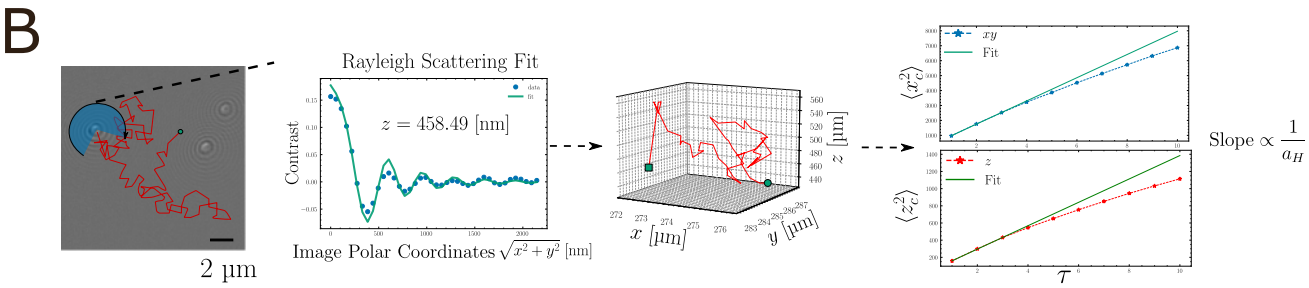


Figure 7.3: method overview.

are further investigated by comparing measurements to a zero-parameter model predicted by the linear-superposition diffusion theory.

7.2.1 Polystyrene PBS series

To investigate the effects on ionic concentration. Polystyrene beads were placed into a series of PBS concentrations. This is shown in Fig. 7.4.

For each plot, 100 nm diameter beads were confined into a 220 nm slit verified using inference fringes. At least 30 beads were tracked in 3 dimensions via iSCAT. The axial position was tracked and binned into histograms. The sizes recovered from a transverse diffusion MSD method and sizes recovered from the axial traveled distance were recovered and compared.

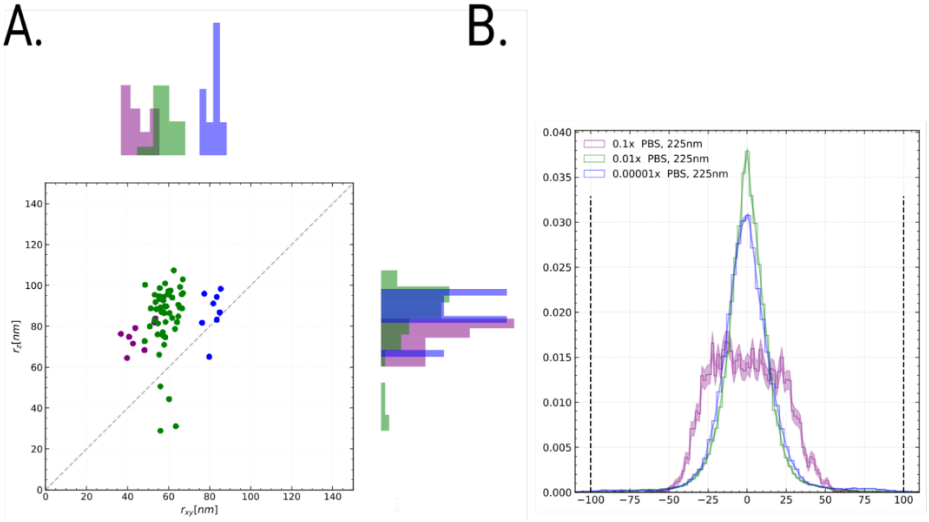


Figure 7.4: Figures Description

7.2.2 Axial Diffusivity

For each plot, the theoretical curve predicted by the linear-superposition theory [1] is presented. Measured single-molecule data is presented overtop this. Axial positions are binned into tracks

and diffusivity is measured on these tracks. This is plotted for single-molecule data and ensemble averages. These plots are shown in Fig. 7.5

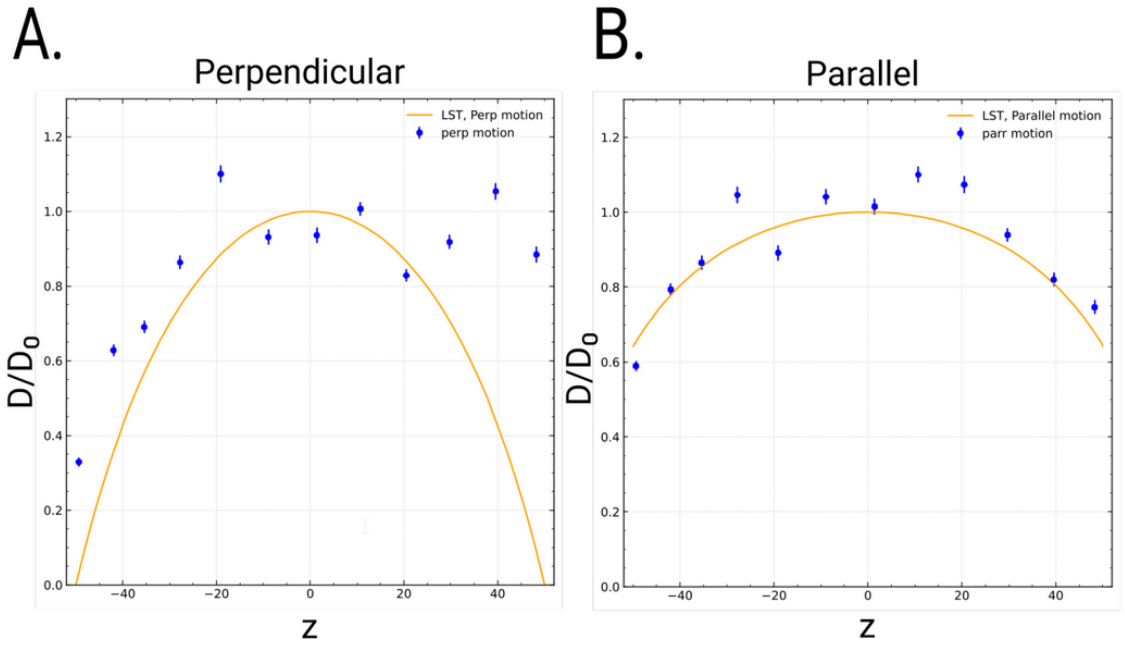


Figure 7.5: Figures Description

7.2.3 LNP Diffusivity

LNP diffusivity is evaluated for various LNP formulations... This is a WIP but the data has been taken already

7.3 Discussion of Results

The discussion starts here

Chapter 8

Conclusion

8.1 Conclusion

8.2 Future outlook

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Appendix A

Surface Charge Estimator Derivation

This is a section for the derivation of the surface charge estimator used in the axial diffusion estimation.

A.1 $\hat{B}$ MLE

The Force used in the modelling is given by;

$$V'(z) = B\frac{k_bT}{\lambda_D} \cdot \exp(-H/\lambda_D) \cdot (\exp(-z_n/\lambda_D) - \exp(z_n/\lambda_D)) \tag{A.1}$$

The dynamics are governed by the equation, written in the Ito convention,

$$\Delta z = D'(z)\Delta t - \beta D(z)V'(z)\Delta t + \sqrt{2D_z}W_t. \tag{A.2}$$

Re-writing in terms of drift and diffusion coefficients;

$$\Delta z = \left[D_z'(z) - \beta BD(z)\phi(z)\right]\Delta t + \sqrt{2D_z}W_t. \tag{A.3}$$

where $\phi(z) = \frac{k_bT}{\lambda_D}\exp(-H/\lambda_D) \cdot (\exp(-z_n/\lambda_D) - \exp(z_n/\lambda_D))$. This is a Markov-process that has a propagator probability from the Onsager-Machlup path integral representation,

$$P(z_{n+1}|z_n) = \frac{1}{4D_z\Delta t}\exp\left[-\frac{\left(\Delta z_n - D_z'(z) + \beta BD(z)\phi(z)\right)^2}{4D_z\Delta t}\right]. \tag{A.4}$$

To obtain an estimator for B, we can take the log-likelihood and differentiate, maximizing it.

$$
\begin{aligned}
l(B) &= \sum_{n=0}^{\infty} \left[-\frac{\left(\Delta z_n - D_z'(z) + \beta B D(z)\phi(z)\right)^2}{4D_z\Delta t} \right] + \text{const} \\
&= -\frac{1}{2}\sum_{n=0}^{\infty} \frac{1}{2D_z\Delta t} \left(\Delta z_n - D_z'(z) + \beta B D(z)\phi(z)\right)^2 + \text{const.}
\end{aligned}
$$

Define $r_n = \Delta z_n - D_z'(z)$ and $w_n = \frac{1}{2D_z}$. Find the maximum of the log-likelihood by taking the derivative of l with respect to B.

$$
l'(B) = -\sum_{n=0}^{\infty} \frac{w_n}{\Delta t} \left(r_n + \beta B D(z)\phi(z)\right) \beta D(z)\phi(z)
$$

We can set this to 0 and solve for $\hat{B}$.

$$
\hat{B} = \frac{\sum_n \frac{\phi(z)}{2D_z} \left[\Delta z - D'(z)\Delta t\right]}{\sum_n \frac{\phi^2(z)\Delta t}{2D_z}} \tag{A.5}
$$

A.2 $\mathbb{V}[\hat{B}]$

The variance of this can be found using the fisher information matrix;

$$
\begin{aligned}
\mathcal{I}(B) &= -l''(B) \\
&= \sum_{n=0}^{\infty} \frac{\Delta t \phi^2(z)}{2D(z)}
\end{aligned}
$$

In the univariate case, the variance is proportional to the inverse of the fisher information, or the denominator of the MLE.

$$
\mathbb{V}[\hat{B}] = \sum_{n=0}^{\infty} \frac{\phi^2(z)\Delta t}{2D(z)} \tag{A.6}
$$

This is an appication of the delta-method as

$$
\begin{aligned}
W_t &\sim \mathcal{N}(0,1) \\
B &= g(W_t).
\end{aligned} \tag{A.7}
$$

Appendix B

Axial Estimation Algorithm Pseudocode

This section of the appendix lists all detailed algorithms and computational methods presented in the axial-estimation section of this project.

The following algorithms are presented:

Function	Desc.
Raw2iScat:	Camera image to iSCAT contrast stack.
TransverseLocalization:	(x, y) localization function with sub-pixel resolution.
RadialIntegral:	Radial integral function.
Nuisance-parameter estimation:	Search algorithm with a grid-scan and raw-python implemented least-squares for Model.
Axial Chain-Link:	Linking algorithm for axial trajectories. Analagous to Munkre's algorithm in 2D tracking.

Table B.1: List of functions presented in appendix. Function names and descriptions are listed.

Each algorithm is presented in language-general mathematical pseudocode. Verbal explanations are described in chapter 6. The appendix serves as a more-detailed explanation so that the interested reader may implement the software themselves and have exact computational details.

A Python implementation, written by the author and open-source contributions, can be produced upon request.

Algorithm 1 Raw image stack → iSCAT contrast stack

Require: $I_{\text{raw}} \in \mathbb{R}^{F\times H\times W}$ ▷ F frames of $H \times W$ pixels
Ensure: $I_{\text{iSCAT}} \in \mathbb{R}^{F\times H\times W}$

Parameters

$R_1 \leftarrow \max(H,W)/100, \quad R_2 \leftarrow \max(H,W)/4$ ▷ band-pass radii in k-space [px]
$\Delta f \leftarrow 25, \quad \Delta L \leftarrow 15$ ▷ temporal gap; background half-window, empirical for data conditions.
$N \leftarrow$ border width ▷ pixels discarded at the edges

Pre-computation (frame independent)

$(k_x, k_y) \leftarrow \textsc{Meshgrid}\big(\big[-\frac{H}{2}, \frac{H}{2}\big), \big[-\frac{W}{2}, \frac{W}{2}\big)\big)$ ▷ centred spatial frequencies
$h \leftarrow \mathbf{1}\big[\, R_1^2 < k_x^2 + k_y^2 < R_2^2 \,\big]$ ▷ annular band-pass filter
$M_{pq} \leftarrow \mathbf{1}[\, N \le p < H - N \ \wedge \ N \le q < W - N \,]$ ▷ edge mask

function $\textsc{AmpNorm}(I, M)$ ▷ row-wise amplitude normalisation of one frame. Row-wise means to avoid readout noise from the camera. Mean over rows to remove line artifacts. (From Vahid's Method)
    $I^{(1)} \leftarrow I \odot M$ ▷ $\odot$: element-wise (Hadamard) product
    $I^{(2)} \leftarrow I^{(1)} / \langle I^{(1)} \rangle$ ▷ $\langle\cdot\rangle$: mean over all pixels
    for each row $p \in \{0, \dots, H-1\}$ **do**
        $a_p \leftarrow \langle M_{p,:} \rangle / \langle I^{(2)}_{p,:} \rangle$ ▷ row means
        $I^{(3)}_{p,:} \leftarrow I^{(2)}_{p,:} / a_p$
    end for
    return $I^{(3)}$
end function

Main loop

for $i = 0$ **to** $F - 1$ **do**
    $I_0 \leftarrow \textsc{AmpNorm}(I_{\text{raw}}[i],\, M)$ ▷ signal frame

    $\mathcal{B}_i \leftarrow \big[\, i - \Delta f - \Delta L,\ i - \Delta f \,\big] \ \cup \ \big[\, i + \Delta f,\ i + \Delta f + \Delta L \,\big]$ ▷ background frame indices
    $\mathcal{B}_i \leftarrow \mathcal{B}_i \cap [\, 0, F \,)$ ▷ clip at the ends of the stack
    $\bar{B} \leftarrow \dfrac{1}{|\mathcal{B}_i|} \displaystyle\sum_{j \in \mathcal{B}_i} \textsc{AmpNorm}(I_{\text{raw}}[j],\, M)$ ▷ mean background
    assert $\bar{B}$ contains no NaN

    $I_m \leftarrow \big(I_0 \oslash \bar{B}\big) \odot M$ ▷ $\oslash$: element-wise division
    $I_{\text{iSCAT}}[i] \leftarrow \mathcal{F}^{-1}\{h \odot \mathcal{F}\{I_m\}\}$ ▷ band-pass filtering
end for

return I_{iSCAT}

Algorithm 2 Transverse Localization algorithm

Inputs
I_{iSCAT} ▷ from Alg. 1
$I_{\text{RVT}} \leftarrow \text{RVT}[I_{\text{iSCAT}}]$

Pre-computation (frame independent)
$t \leftarrow 0.9 \cdot \max I_{\text{iSCAT}}$ ▷ threshold
$n \leftarrow 30$ ▷ number of particles per frame to assume, empirical
$x_{\text{ROI}}, y_{\text{ROI}} \leftarrow 5$ ▷ ROI cropping for radial symmetry algorithm
Fitting Routine
for $i = 0$ **to** $F - 1$ **do**
    $I_{\text{dil}} \leftarrow \textsc{ImDialate}(I_{\text{RVT}}[i])$
    $\{x, y\} \leftarrow \textsc{Where}\,[I_{\text{dil}} == I_{\text{RVT}} \wedge I_{\text{RVT}} > t]$ ▷ Candidate Positions
    $\{x_1, y_1\}|_n = \textsc{GaussianFit}(x, y)$ ▷ Non-linear least squares fitting, take best n fits
    $\{x_\star, y_\star\} = \textsc{RadialSymmetry}(x_1, y_1, x_{\text{ROI}}, y_{\text{ROI}})$ ▷ from [27]
end for
2D Trace Linking
Traces $\leftarrow \textsc{Munkres}[x_\star, y_\star]$ ▷ Minimization of distance cost-matrix
ROI $= I_{\text{iSCAT}}[Traces]$ ▷ Cropping of image around every frame for each trace.

Algorithm 3 Discrete radial profile (angular average over a wedge)

Require: $I \in \mathbb{R}^{F\times H\times W}$ ▷ stack of F image ROIs
Require: $\mathbf{c} = (c_p, c_q)$ ▷ ROI centre (may be sub-pixel)
Require: θ_0, $\Delta\theta$ ▷ start angle and angular width of the wedge
Require: R ▷ outer radial bound, in pixels
Ensure: $g \in \mathbb{R}^{F\times R}$, where

$$g_t(\rho) \;\simeq\; \frac{1}{\Delta\theta}\int_{\theta_0}^{\theta_0+\Delta\theta} I_t(\rho,\theta)\,\mathrm{d}\theta$$

is the angular mean of frame t over the wedge, as a function of radius.

Pre-computation (frame independent)

$\rho_{pq} \leftarrow \big\|(p,q)-\mathbf{c}\big\|_2, \quad \theta_{pq} \leftarrow \operatorname{arctan2}\big(p-c_p,\; q-c_q\big) \bmod 2\pi$ ▷ polar coordinates of each pixel

for $r = 0$ **to** $R-1$ **do** ▷ annular-wedge bins
  $\mathcal{A}_r \leftarrow \big\{\,(p,q)\;:\; r \le \rho_{pq} < r+1 \;\wedge\; \big(\theta_{pq}-\theta_0\big) \bmod 2\pi \;<\; \Delta\theta\,\big\}$
  $N_r \leftarrow |\mathcal{A}_r|$ ▷ $N_r \approx (r+\frac{1}{2})\,\Delta\theta$
end for

Main loop

for $t = 0$ **to** $F-1$ **do**
  for $r = 0$ **to** $R-1$ **do**
    if $N_r = 0$ **then**
      $g_t(r) \leftarrow \textsc{NaN}$ ▷ empty bin: undefined, not zero
    else
      $g_t(r) \leftarrow \dfrac{1}{N_r}\displaystyle\sum_{(p,q)\in\mathcal{A}_r} I_t[p,q]$ ▷ angular average at radius r
    end if
  end for
end for

return g

Algorithm 4 Nuisance-parameter estimation by profiled grid search + local refinement

Require: $g \in \mathbb{R}^M$ ▷ radial profile, sampled at radii r_i, $i = 1, \dots, M$
Require: $r \in \mathbb{R}^M$, $w \in \mathbb{R}^M_{\geq 0}$ ▷ radii [nm] and fit weights
Require: $k = 2\pi n_0/\lambda$ ▷ optical wave-vector in the medium
Require: $[z_-, z_+]$, $[E_-, E_+]$, $[\gamma_-, \gamma_+]$ ▷ parameter bounds
Require: $\Delta_\lambda = 32$

iPSF Model

$G_i(z, \gamma) \;=\; \frac{q(r_i,z)}{\varrho(r_i,z)^2} \cdot \cos\big[\, k\,(z + \varrho(r_i, z))\,\big]\, e^{-r_i/\gamma}, \quad \varrho(r, z) = \sqrt{r^2 + z^2}$

$\chi^2(E_0, z, \gamma) \;=\; \big\langle\, g - E_0 G,\; g - E_0 G \,\big\rangle_w, \quad \langle a, b\rangle_w \;\equiv\; \sum_{i=1}^{M} w_i\, a_i\, b_i$

Pre-computation

$\delta_z \leftarrow 2\pi/(k\,\Delta_\lambda)$ ▷ $= (\lambda/n_0)/\Delta_\lambda$: fringe period / Δ_λ

$\mathcal{Z} \leftarrow \{\, z_- + m\,\delta_z \;:\; m = 0, \dots, N_z - 1 \,\}, \quad N_z = \lceil (z_+ - z_-)/\delta_z \rceil$

$\Gamma \leftarrow \{\, \gamma_1, \dots, \gamma_{N_\gamma} \,\} \subset [\gamma_-, \gamma_+]$ ▷ decay-length grid

$\langle g, g\rangle_w \leftarrow \sum_i w_i\, g_i^2$

Profiled grid search

for $\gamma \in \Gamma$ **do**
  for $z \in \mathcal{Z}$ **do**
    $\varrho_i \leftarrow \sqrt{r_i^2 + z^2} \quad \forall\, i$
    $G_i \leftarrow \dfrac{q(r_i, z)}{\varrho_i^2}\, \cos\big[k(z + \varrho_i)\big]\, e^{-r_i/\gamma} \quad \forall\, i$
    $\langle G, G\rangle_w \leftarrow \sum_i w_i\, G_i^2, \quad \langle G, g\rangle_w \leftarrow \sum_i w_i\, G_i\, g_i$
    $\rho(z, \gamma) \leftarrow \dfrac{\langle G, g\rangle_w}{\sqrt{\langle G, G\rangle_w\, \langle g, g\rangle_w \;+\; \varepsilon}}$ ▷ weighted correlation coefficient
    $\hat{E}_0(z, \gamma) \leftarrow \textsc{Clip}\left[\, \dfrac{\langle G, g\rangle_w}{\langle G, G\rangle_w},\; E_-,\; E_+ \right]$ ▷ profiled amplitude
    $\mathcal{L}(z, \gamma) \leftarrow 1 - \rho^2(z, \gamma)$ ▷ $\chi^2_{\min}/\langle g, g\rangle_w$
  end for
end for

Refinement

$(z^\star, \gamma^\star) \leftarrow \underset{(z,\gamma)\,\in\,\mathcal{Z}\times\Gamma}{\arg\min}\; \mathcal{L}(z, \gamma)$ ▷ coarse global minimum

$E_0^\star \leftarrow \hat{E}_0(z^\star, \gamma^\star)$

$(\hat{E}_0, \hat{z}, \hat{\gamma}) \leftarrow \text{L-BFGS-B}\big[\, \sum_i \mathcal{H}_\delta\big(g_i - E_0 G_i(z, \gamma)\big);\ \text{init } (E_0^\star, z^\star, \gamma^\star);\ \text{bounds } [E_\pm], [z_\pm], [\gamma_\pm] \,\big]$
▷ Huber loss $\mathcal{H}_\delta$

return $(\hat{E}_0,\, \hat{z},\, \hat{\gamma})$

Algorithm 5 Dynamic program for the optimal 2D chain of candidate points, Viterbi Algorithm

Require: Candidate sets $\mathcal{C}_t = \{\, \mathbf{p}_t^{(i)} = \big(x_t^{(i)},\, y_t^{(i)}\big) \,\}_{i=1}^{n_t},\ t = 0, \dots, F-1$
where $x_t^{(i)} \equiv \texttt{min_idx}[t][i]$ (position) and $y_t^{(i)} \equiv \texttt{min_vals}[t][i]$ (value)
Require: Weights $\mathbf{W} = \mathrm{diag}(w_x, w_y)$, default $\mathbf{W} = \mathbf{I}$
Require: Maximum step $s_{\max} \in (0, \infty]$ ▷ $s_{\max} = \infty \Rightarrow$ unconstrained
Ensure: Chain $\mathbf{k} = (k_0, \dots, k_{F-1})$, $k_t \in \{1, \dots, n_t\}$, minimizing

$$J(\mathbf{k}) \;=\; \sum_{t=1}^{F-1} \big\| \mathbf{W}\big(\mathbf{p}_{t-1}^{(k_{t-1})} - \mathbf{p}_t^{(k_t)}\big) \big\|_2 \qquad \text{s.t.} \quad \big\| \mathbf{W}\big(\mathbf{p}_{t-1}^{(k_{t-1})} - \mathbf{p}_t^{(k_t)}\big) \big\|_2 \le s_{\max}$$

together with the corresponding x- and y-chains.

Forward pass ▷ $C_t(j)$: cost of the best chain ending at $\mathbf{p}_t^{(j)}$

$C_0(i) \leftarrow 0 \quad \forall\, i \in \{1, \dots, n_0\}$
for $t = 1$ **to** $F-1$ **do**
▷ Weighted Euclidean transition cost, $n_{t-1} \times n_t$
  $d_t(i,j) \leftarrow \sqrt{w_x^2\big(x_{t-1}^{(i)} - x_t^{(j)}\big)^2 + w_y^2\big(y_{t-1}^{(i)} - y_t^{(j)}\big)^2} \quad \forall\,(i,j)$
  $d_t(i,j) \leftarrow \infty$ **if** $d_t(i,j) > s_{\max}$ ▷ forbid over-long steps

▷ Bellman recursion
  $\pi_t(j) \leftarrow \arg\min_i \big[\, C_{t-1}(i) + d_t(i,j) \,\big] \quad \forall\, j \in \{1, \dots, n_t\}$
  $C_t(j) \leftarrow C_{t-1}\big(\pi_t(j)\big) + d_t\big(\pi_t(j),\, j\big)$

  if $C_t(j) = \infty \ \forall\, j$ **then**
    return $\varnothing$ ▷ no admissible chain survives frame t
  end if
end for

Backtracking
$k_{F-1} \leftarrow \arg\min_j C_{F-1}(j)$ ▷ cheapest terminal node
for $t = F-1$ **down to** 1 **do**
  $k_{t-1} \leftarrow \pi_t(k_t)$
end for

return $\big\{k_t\big\}, \quad \big\{x_t^{(k_t)}\big\}, \quad \big\{y_t^{(k_t)}\big\} \quad$ for $t = 0, \dots, F-1$